# **Employment Status and Credit Cycles:** An Extensive Margin Analysis in Heterogeneous Agent Life Cycle Models.

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# Motivation

**Motivation:**

- The Global Financial Crisis demonstrated that unemployment and household debt deleveraging occur together.
- Loan demand falls because the youth are the main borrowers and face the steepest rise in unemployment.

| | Greece | Spain | Portugal | Italy |
|---|---|---|---|---|
| Unemployment ($\Delta$ pp) | +20.5 | +18.3 | +9.2 | +6.9 |
| Youth unemployment ($\Delta$ pp) | +41.2 | +39.0 | +24.2 | +23.6 |
| Household debt (%) | −30.9 | −32.4 | −27.4 | −8.3 |

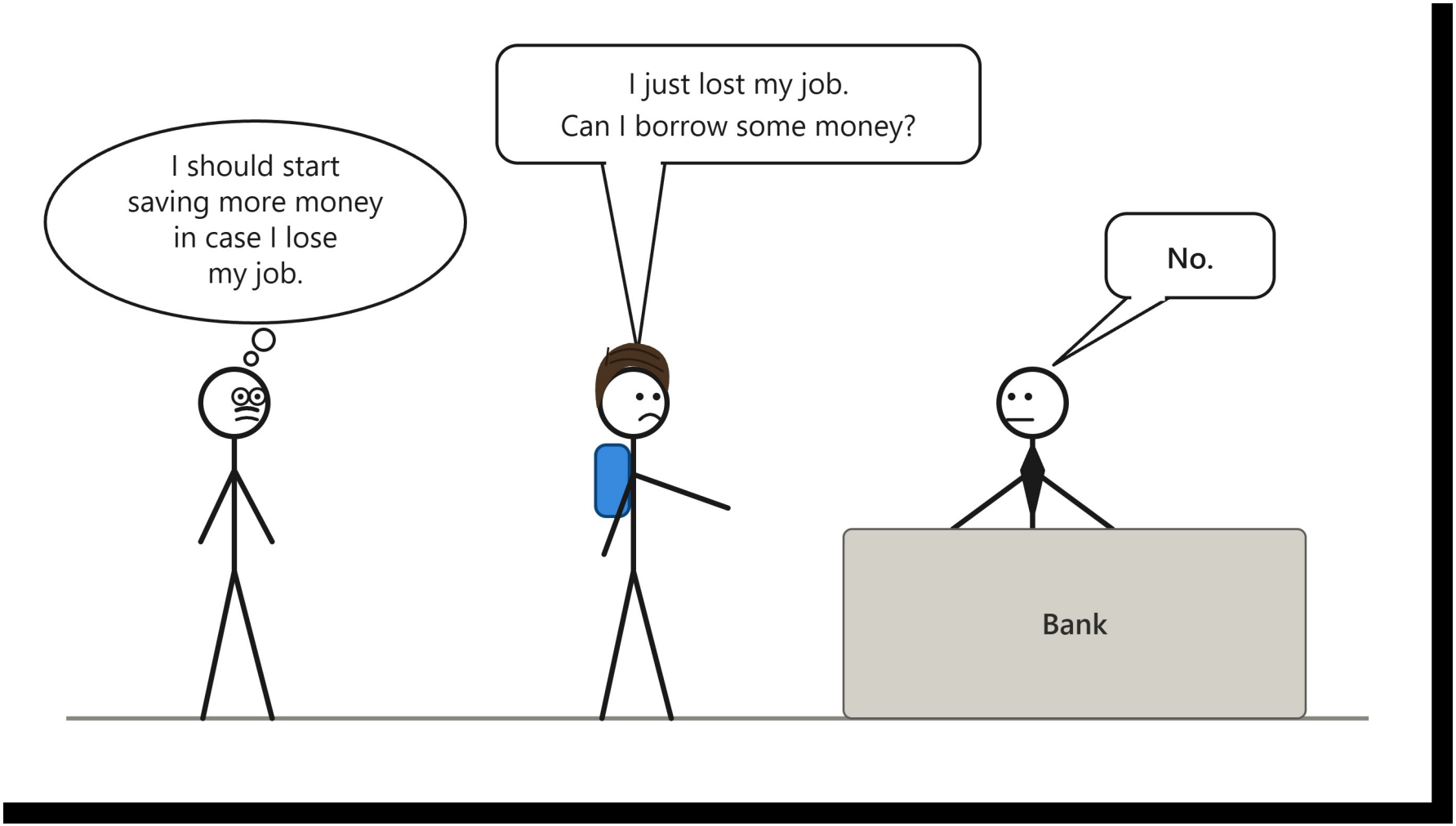
- Standard models miss this: RBC/NK use representative agents, and HANK models adjust labor on the intensive margin, ruling out endogenous tightening of employment based borrowing constraints.

# Research Question

**Research Question:**

- How does introducing unemployment risk modeled as a shock that simultaneously removes labor income and restricts access to credit reshape household borrowing, saving, and the loanable funds market?

# Research Question Simplified




*Notes*: How does this situation affect borrowing, saving, and the loanable funds market?

# Addressing the Gap

**Captures Unemployment's "Double Shock":**

- Unemployed lose income and access to credit because borrowing is tied to employment status.

**Intergenerational Lending (OLG):**

- Young borrow from middle aged savers, matching real world dynamics

**Labor Adjusts on Both Margins:**

- Firms decide effective labor levels that fail to align with the utility maximizing preferences of the household
- Extensive (headcount) vs. intensive (hours) margins compared.

**What This Allows Us To See:**

- How labor dynamics drive saving, consumption, and the loanable funds market.
- Policy analysis of how counter cyclical transfer policies affect the model.

# Roadmap

**Where We Are Going:**

- **Literature**
- **Motivating Data**
- **OLG Benchmark**
- **Heterogeneous Agent Model**
- **HA Model IRF**
- **Empirical IRF**

# Heterogeneous Agents, OLG, and Labor Flows

**Kaplan & Violante, (2018):**

- Adding household heterogeneity from idiosyncratic productivity shocks (RA vs. HA, up to the two-asset HANK) changes the macro response to aggregate shocks.
- "The fact that labor market risk is mostly exogenous in current HANK models is a gaping hole in the framework."
- I build directly on this HA structure.

**Eggertsson, Mehrotra, & Robbins (2019):**

- Used an OLG endowment economy of secular stagnation.
- Introduced youth unemployment to their 3 agent OLG model.

**Shimer (2012):**

- In a continuous-time framework, movements in the job-finding probability account for most unemployment fluctuations (77% since 1948, 90% after 1987).
- These labor market dynamics are used to discipline our model.

# Data used

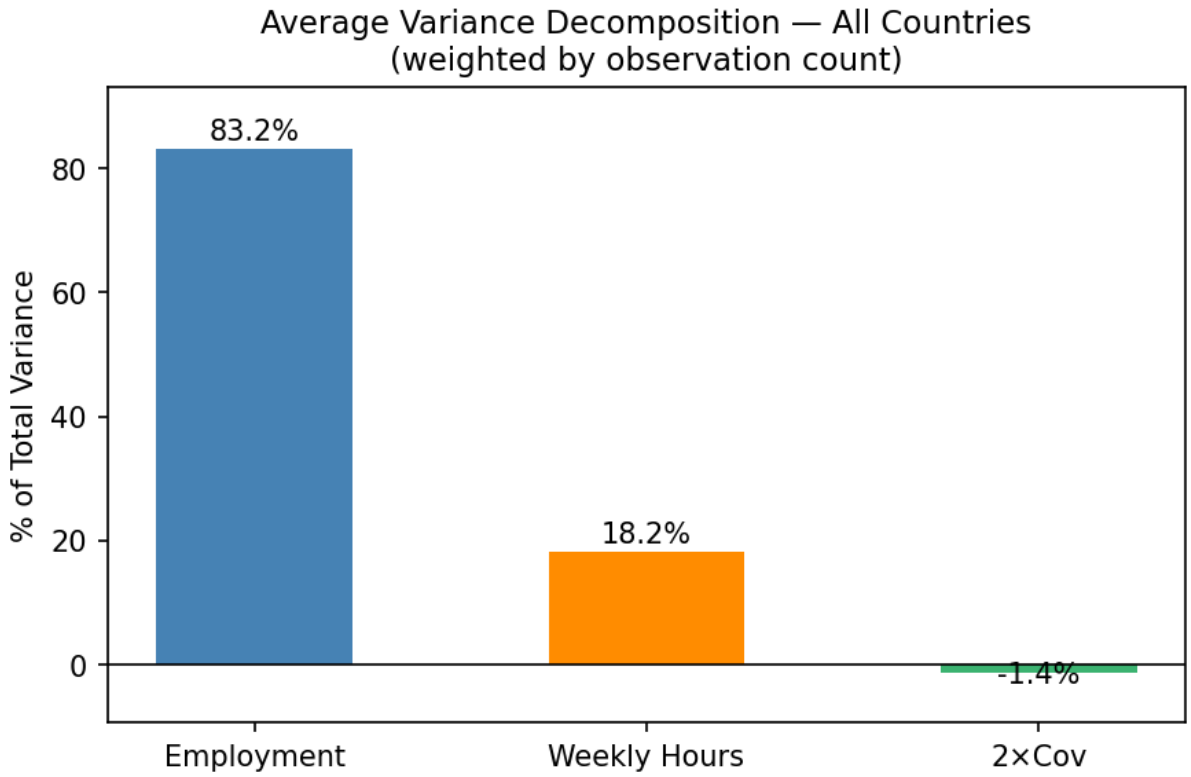
**US Data:**

- Data is Quarterly from 1952 to 2019
- US lacks state level household debt data.

**European Data As a Complement:**

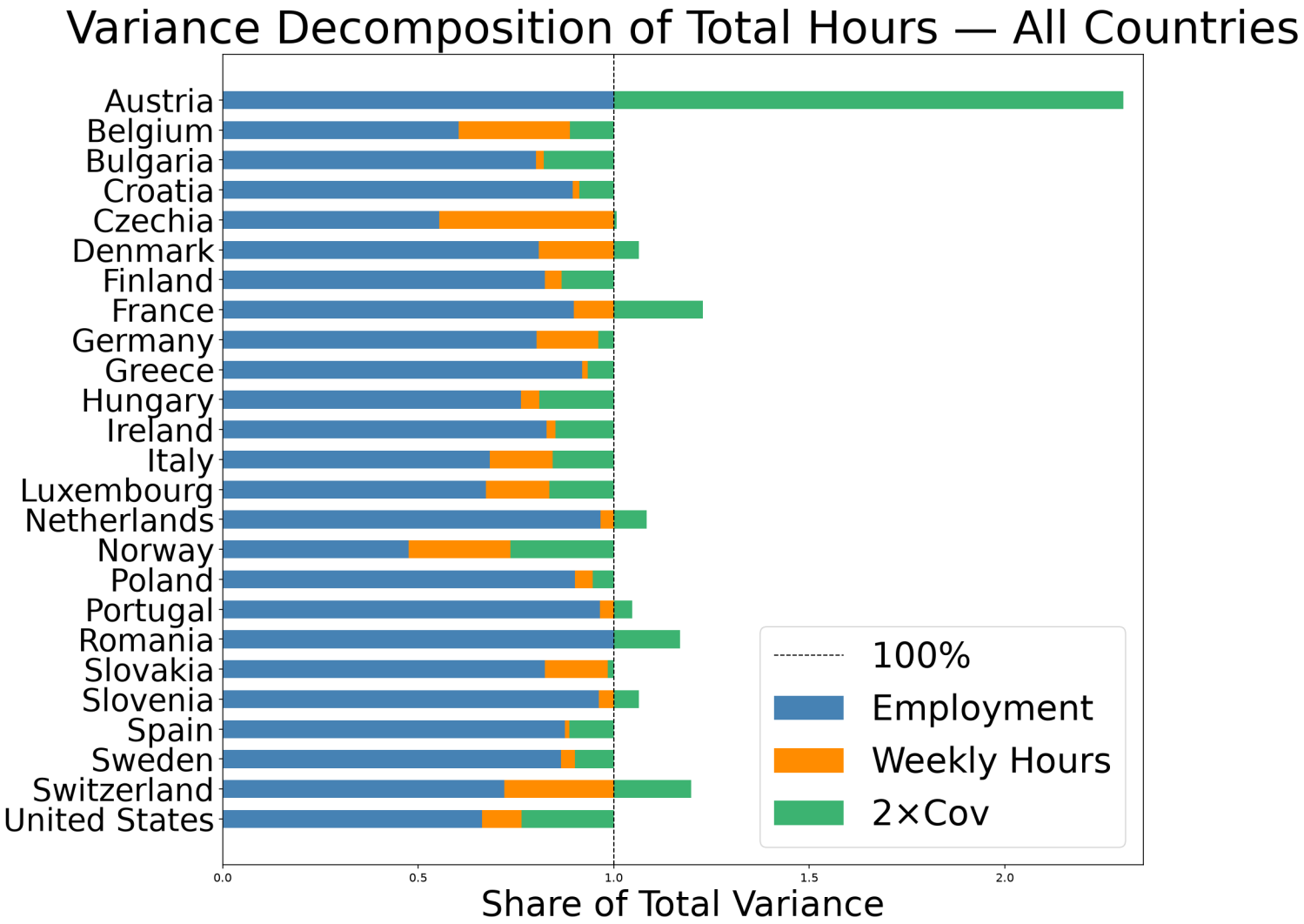
- Data is Quarterly from 1996 to 2019
- Greater cross country variation for identification.
- High capital mobility means European household leverage mainly reflects country specific loan demand.
- Europe's demographic cycle leads the US by 10–20 years.
- Southern Europe has faced debt crises, a risk the US increasingly confronts.

# Variance Decomposition

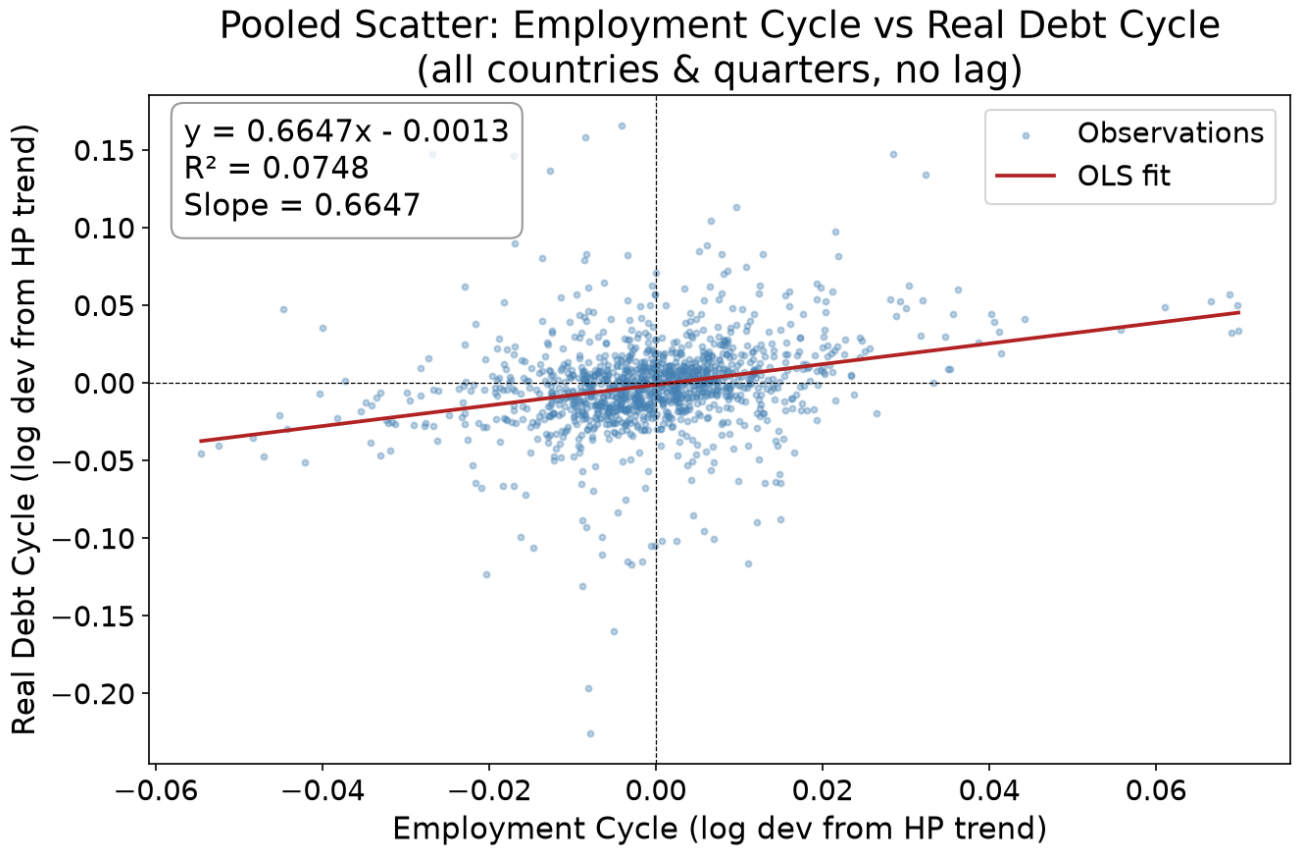



- Employment dominates total variance.
- correlation is barely negative because of Austria's negative covariance.

# Variance Decomposition Within Country



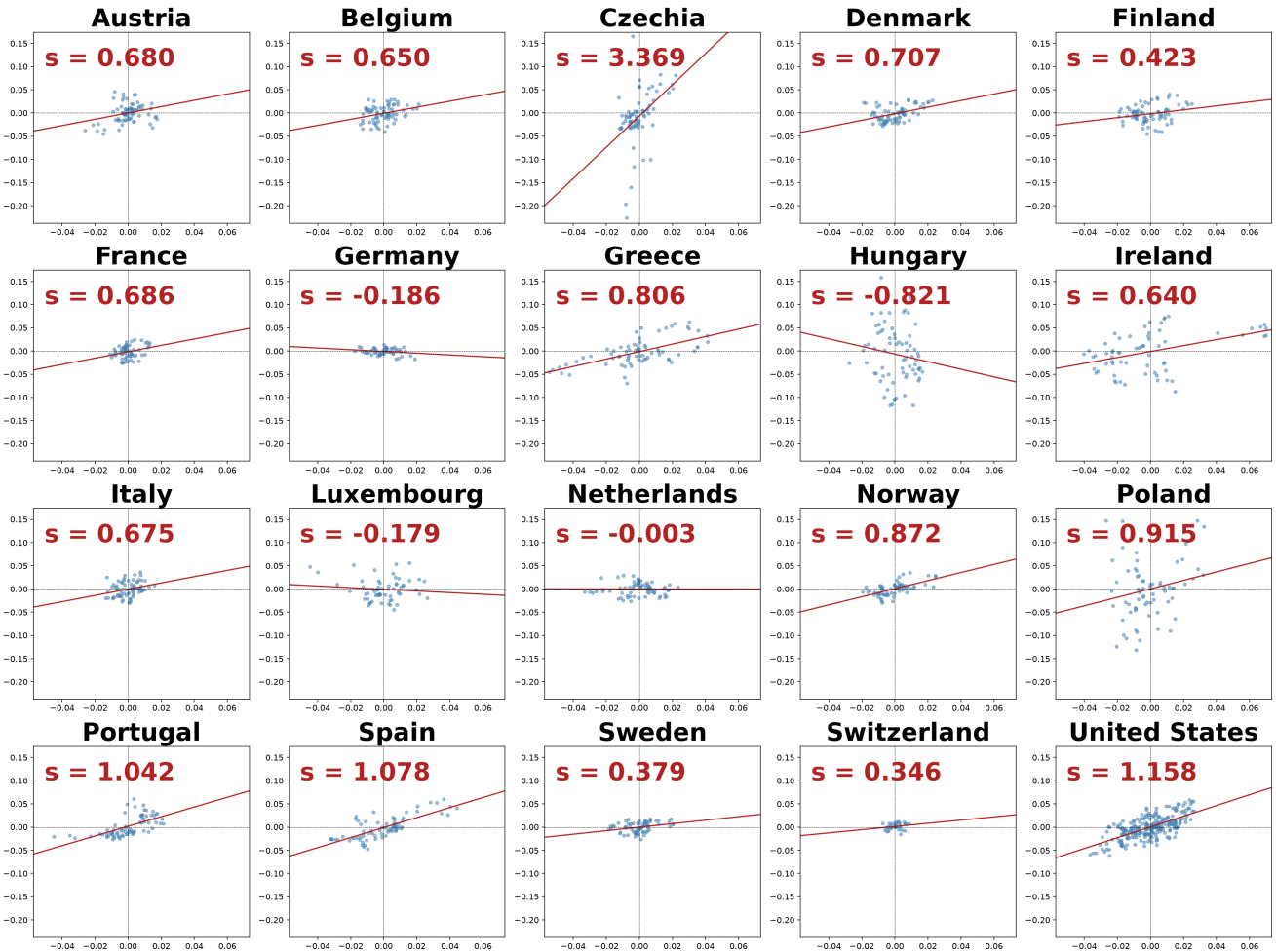
# Employment vs Debt Aggregate




- There is a strong correlation between real detrended household debt and detrended employment.
- Detrending method follows the same framework as Hansen 1985

# Employment vs Debt Country Specific

## Employment Cycle vs Real Debt Cycle — By Country



# OLG Overview

**Key Details**

- Consists of 3 agent types: young, middle aged, and old
- The young agents will be employed or unemployed, highlighting the employment risk.
- An exogenous youth unemployment shock will decrease loan demand and affect the loanable funds market.

# OLG Dynamics

**Key Details:**

| | Income | Borrowing/Dissaving | Consumption |
|---|---|---|---|
| Young Unemployed | + | 0 | + |
| Young Employed | ++ | + | +++ |
| Middle Age Employed | +++++ | − − | +++ |
| Old Retired | ++ | + | +++ |

**Borrowing**

- There is a lifecycle consumption smoothing dynamic
- Borrowing constraint is based on income.
- Borrowing constraint is an exterior solution.

# OLG Setup

- **Utility Function**

$$U_t^y = ln(C_t^y) + \beta ln(C_{t+1}^m) + \beta^2 ln(C_{t+2}^o)$$

- **Debt Constraint**

$$D_t = B_t(1 + r_t) = \gamma Y_t^y$$

- **Euler Equation**

$$\frac{1}{C_t^m} = \beta E_t(1 + r_t)\frac{1}{C_{t+1}^o}$$

# OLG Setup cont.

- **Loan Demand**

$$L^d = \frac{(1-\omega_t)D^e + \omega_t D^u}{1+r_t} = \frac{(1-\omega_t)D^e}{1+r_t}$$

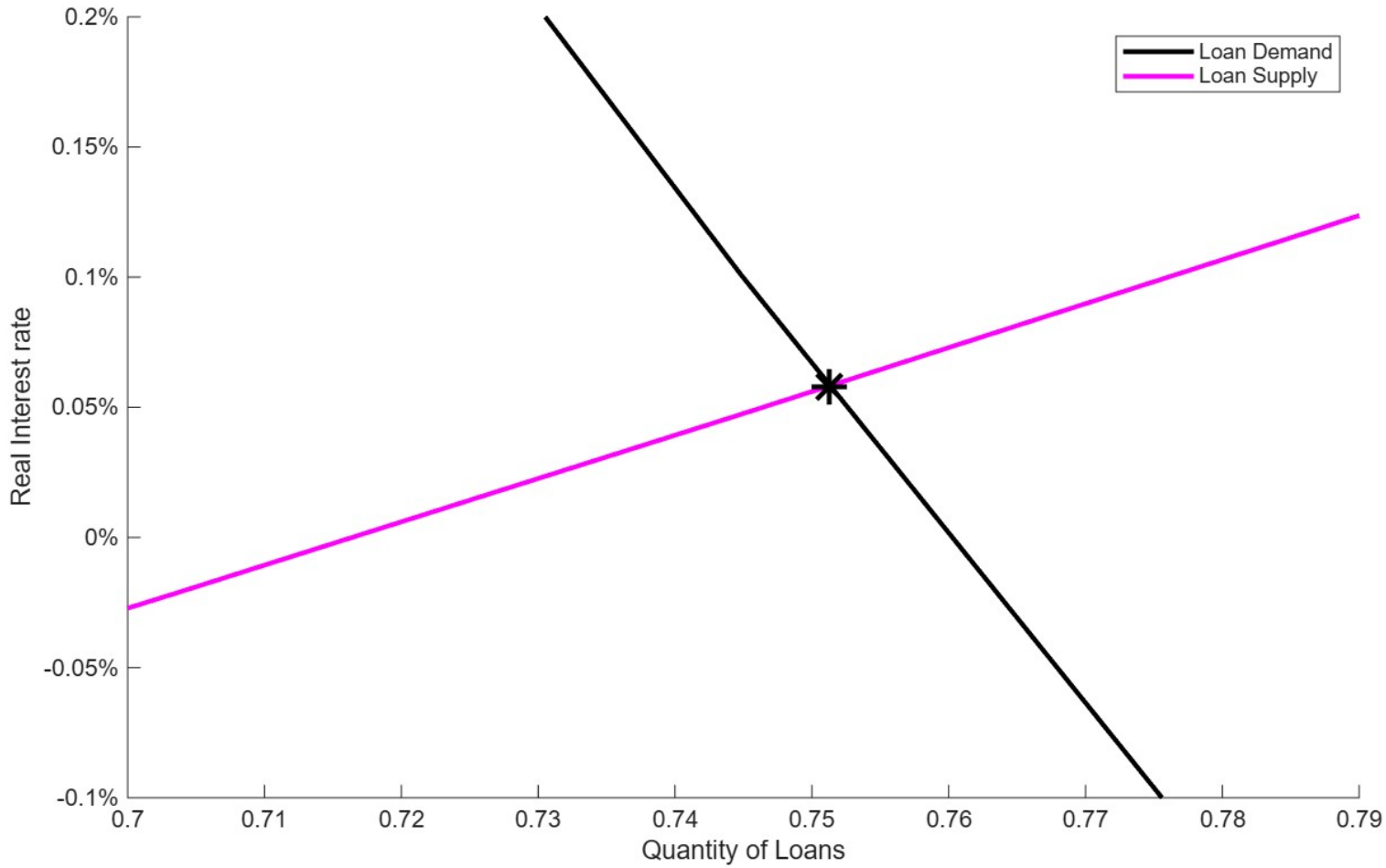
- **Loan Supply**

$$L^s = \left(\frac{\beta}{1+\beta}\right)(Y_t^m - (1-\omega_{t-1})D^e) - \left(\frac{1}{1+\beta}\right)\frac{Y_{t+1}^o}{1+r_t}$$

- **Interest Rate**

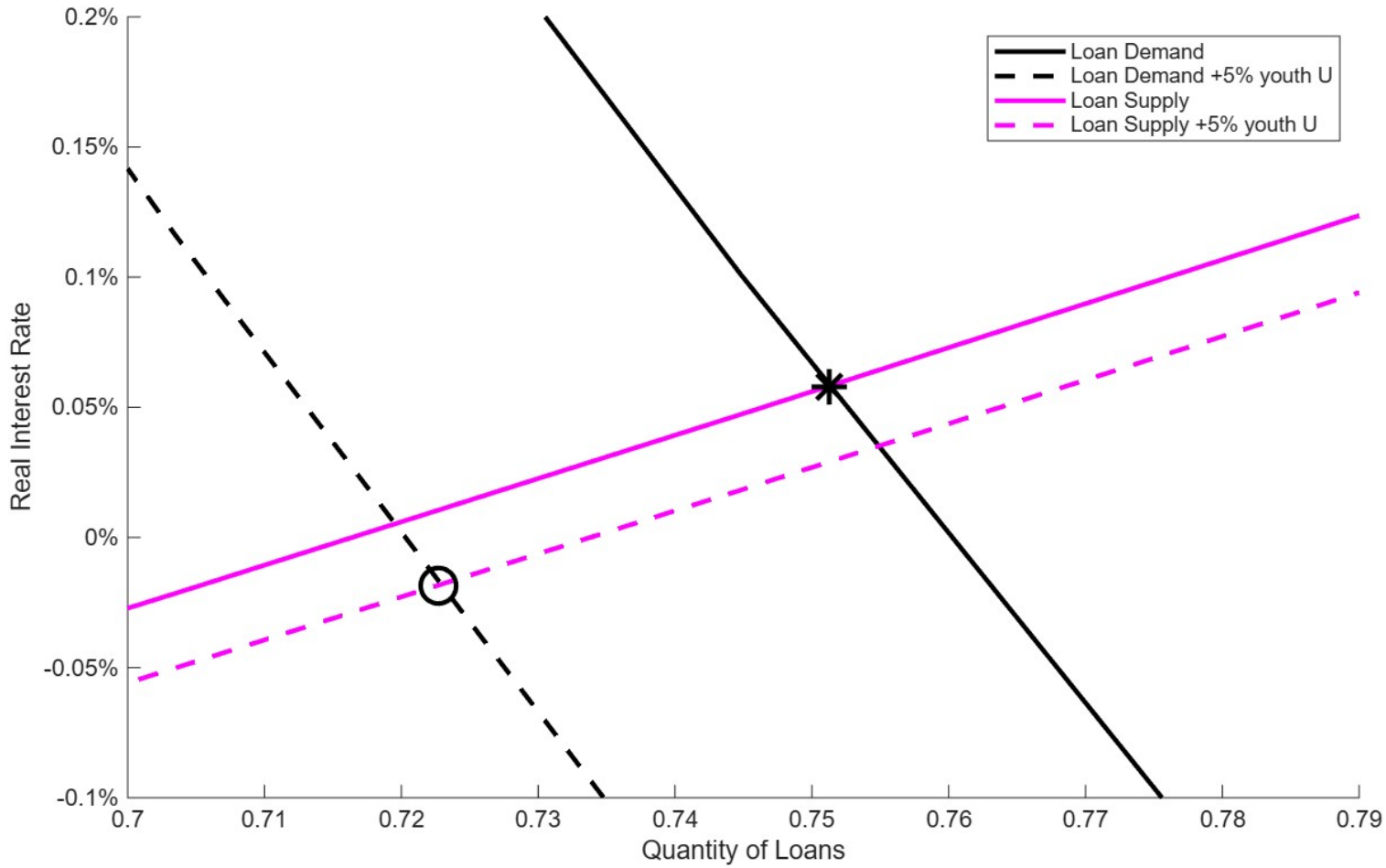
$$1+r_t = \frac{(1+\beta)(1-\omega_t)D^e + Y_{t+1}^o}{\beta(Y_t^m - (1-\omega_{t-1})D^e}$$

# Results: Youth Unemployment's Comparative Static



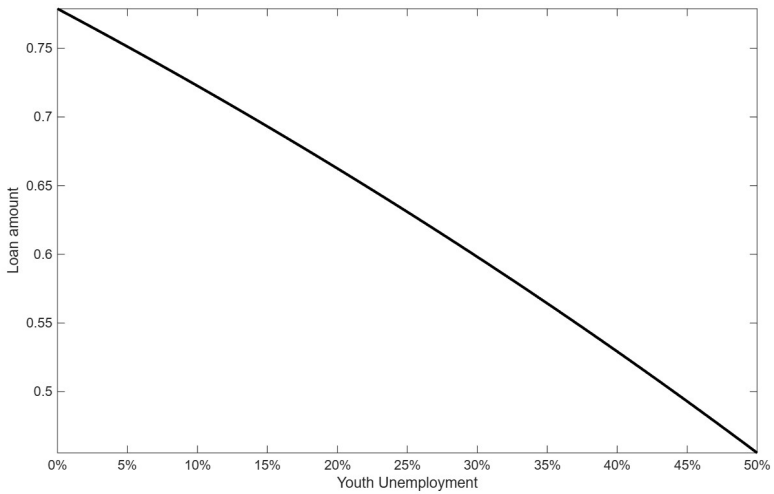

*Before: starting loanable funds market*

# Results: Youth Unemployment's Comparative Static

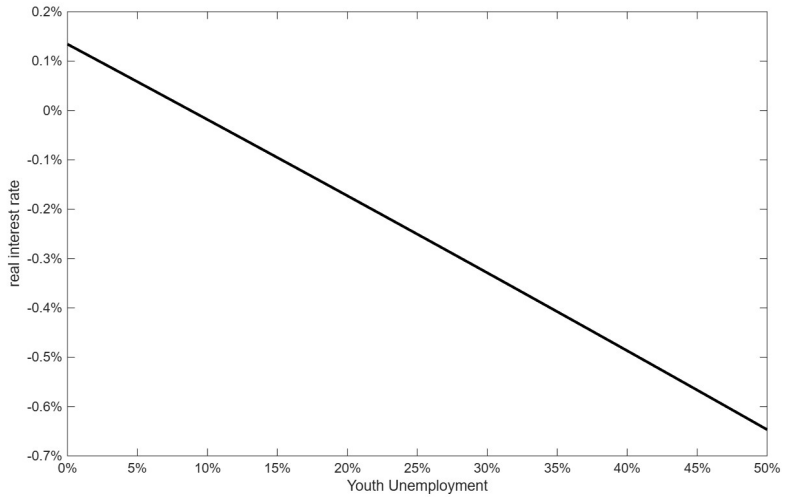



*After: loanable funds market after youth unemployment shock.*

# Results: Youth Unemployment on Total Loans and Interest Rate




*Notes*: The loan amount drops at higher employment rates.




*Notes*: Interest rates drop as youth unemployment goes up.

# HA Overview

**Goal of HA model**

- Move beyond OLG toward a more empirically grounded heterogeneous agent lifecycle model that allows for loan supply to be studied.

**Key Additions:**

- Continuous time modeling
- General equilibrium model with a firm setting the effective labor level based on productivity level.
- Life cycle income dynamics.
- Employment transition dynamics based on Shimer (2012).

# HA Setup

**Key Details:**

- The distribution of agents is based on asset level, employment status (employed/unemployed), and age (20 - 70 w/ death at 75).
- Income is set by age and employment status
- Unemployment/Social Security is .33x of the average wage income in equilibrium.
- The unemployed cannot borrow, but must maintain existing payments out of their unemployment payments.
- There is a precautionary savings motive to hold positive assets. This allows agents to smooth consumption when they are unemployed.

# HA Setup continued

- **Value Function**

$$\rho v(a,e,l) = \max_c \; u(c) + v_a'(a)\dot{a} + \sum_{e' \neq e} \lambda_{ee'}[v(a,e',l) - v(a,e,l)]$$

$$+ \sum_{l' \neq l} \lambda_{ll'}[v(a,e,l') - v(a,e,l)]$$

- **Budget Constraint**

$$c(a,e,l) = (wz_l h_e + u_e(1-e) + \pi_l)(1-\tau) + ra - s(a,e,l)$$

- **Kolmogorov Forward Equation**

$$\partial_t g_t(a,e,l) = \partial_a[s(a,e,l)g_t(a,e,l)] - g_t(a,e,l)\sum_{e' \in E} \lambda_{ee'} + \sum_{x \in E} \lambda_{xe} g_t(a,x,l)$$

$$- g_t(a,e,l)\sum_{l' \in L} \lambda_{ll'} + \sum_{x \in L} \lambda_{xl} g_t(a,e,x)t$$

# Market Clearing Conditions

- **Asset Market Clearing Condition**

$$A(r) = 0 = \sum_e \int_l \int_a ag(a, e, l; r)\, da$$

- **Government Budget Clearing Condition**

$$gov_surplus = 0 = \sum_e \int_l \int_a \tau_1(w(z_l h_{e,t}) + \pi_l)$$

$$- \sum_e \int_l \int_a u_e(1 - e)$$

# Borrowing Constraint

- The amount of debt that can be taken out (negative asset level) is based on a payment to income ratio and a minimum debt payment that amortizes at retirement age.
- Payment to income PTI based on Quercia, R. G., McCarthy, G. W., & Wachter, S. M. (2003) value of .28.

$$pmt_t = .28 y_{e,l} e_e$$

- Minimum Payment amount

$$pmt_t = \frac{-a * r}{1 - (1 + r)^{65-l}}$$

- Assets at the time of death will be given to the new group of 20-year-olds as a bequest from dying grandparents to grandchildren.

# Life Cycle Dynamics

- The income is based on the income by age in the 2013 report by the BLS. The data was fit with a parabola for simplicity.
- The dividend is based on the same methodology. All employed individuals get the same dividend income by age.
- There is forced retirement at age 65. Everyone provides a perfectly inelastic labor supply when below age 65.

# Firm Dynamics

- Effective labor used is the sum of on the productivity by age $z_l$ times hours worked by all agents $h_{e,t}$.

$$N_t = \sum_e \int_l (z_l h_{e,t})$$

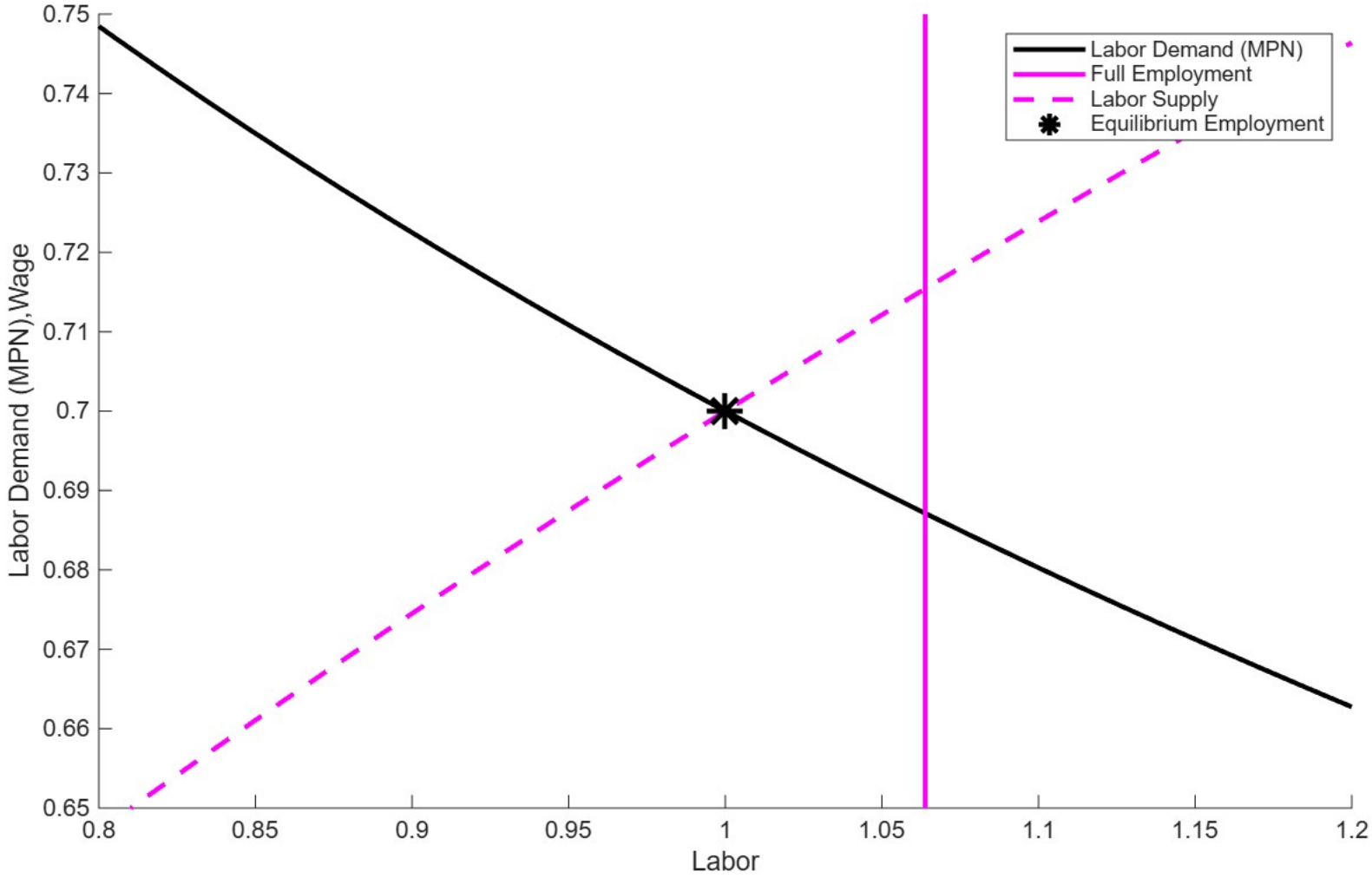
- Standard profit function exists

$$\pi_t = A_t (N_t)^{1-\alpha} - w_t N_t$$

- Quantity of Labor demanded is determined by

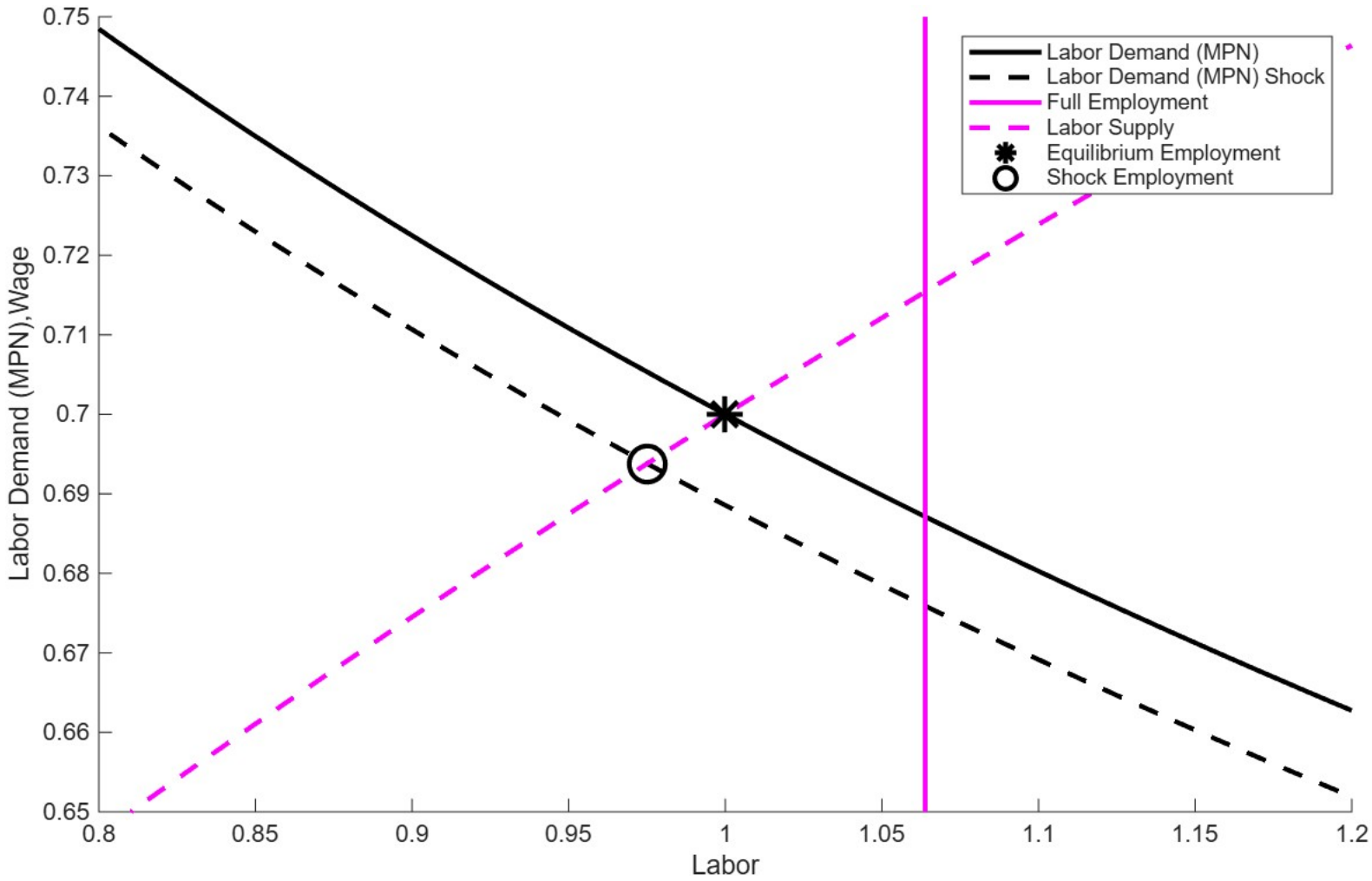
$$N_t = \left(\frac{A_t(1-\alpha)}{w}\right)^{\frac{1}{\alpha}} = A_t^{\frac{1}{\phi+\alpha}}$$

# Labor Market: Before vs. After the Shock




*Notes*: There is unemployment in equilibrium.

# Labor Market: Before vs. After the Shock




*Notes*: After a negative productivity shock drives unemployment by decreasing MPN. This is a departure from when agents choose their hours.

# Employment Dynamics

**Employment Flows (Shimer 2012)**

- From 1948 to 2010, the quarterly average of monthly hiring rates is 44 %. This corresponds to a 82.4 % quarterly hire rate.

**Unemployment Dynamics used to enrich the model**

- The unemployment of 20-24 year olds is 1.7x total unemployment.
- 25-34 unemployment roughly matches the total unemployment rate.
- The separation rate is set to ensure that total unemployment matches the natural rate.
- The chance that 20 and 25 year olds exit employment is set to be higher.
- A portion of 20 year olds will start employment by taking some of the jobs vacated from the retiring 65 year olds.

# Intensive vs Extensive Margin vs Huggett Model

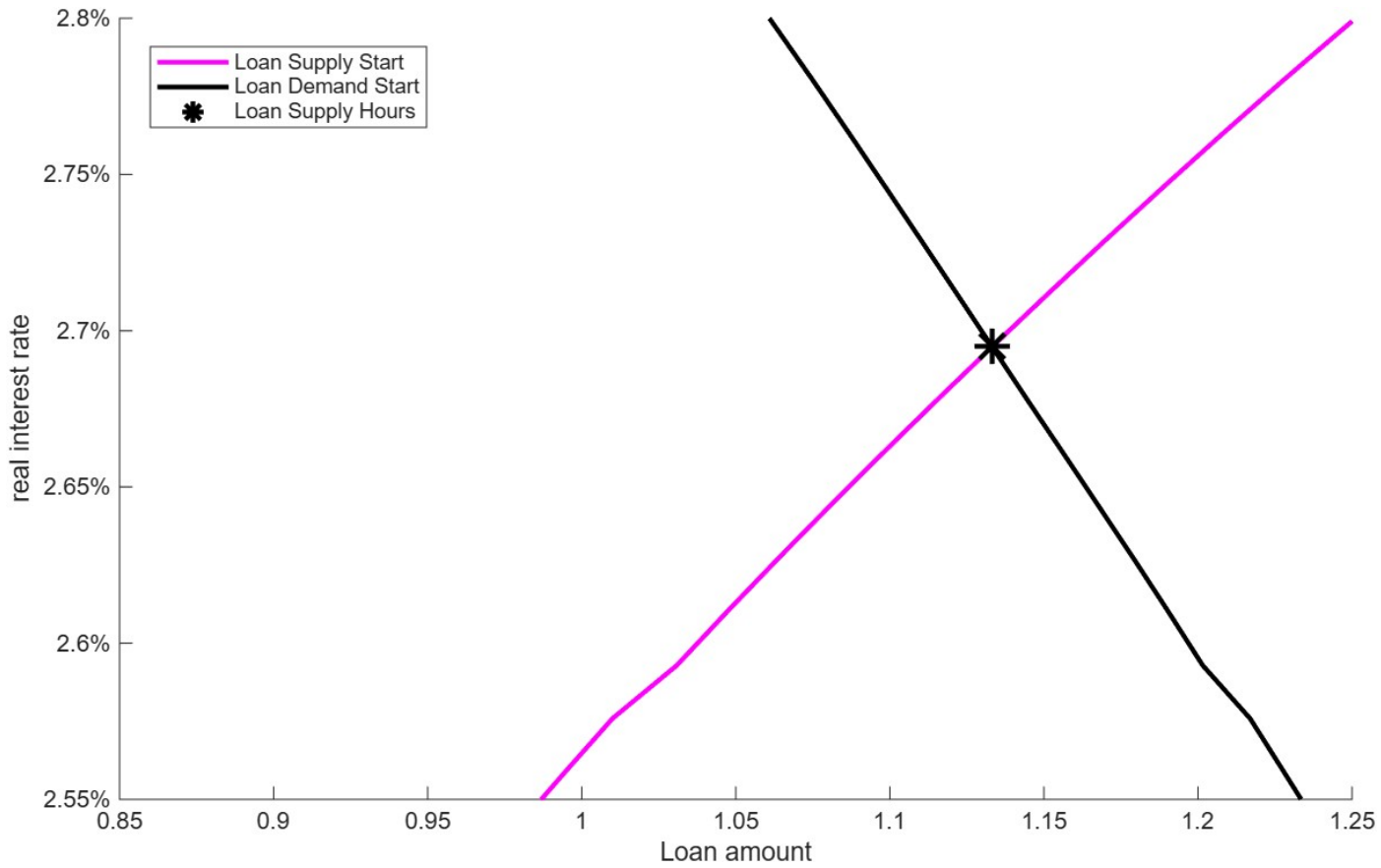
**Intensive vs Extensive Margin**

- After a productivity shock, total hours worked drop.
- The two labor margins (extensive vs intensive) will be compared.
- This allows us to highlight differences between the two margins.

**Modified Huggett Model**

- The modified Huggett model is used to demonstrate that it fails to capture similar dynamics in the loanable funds market.
- Huggett model doesn't have significant changes in the precautionary savings motive.

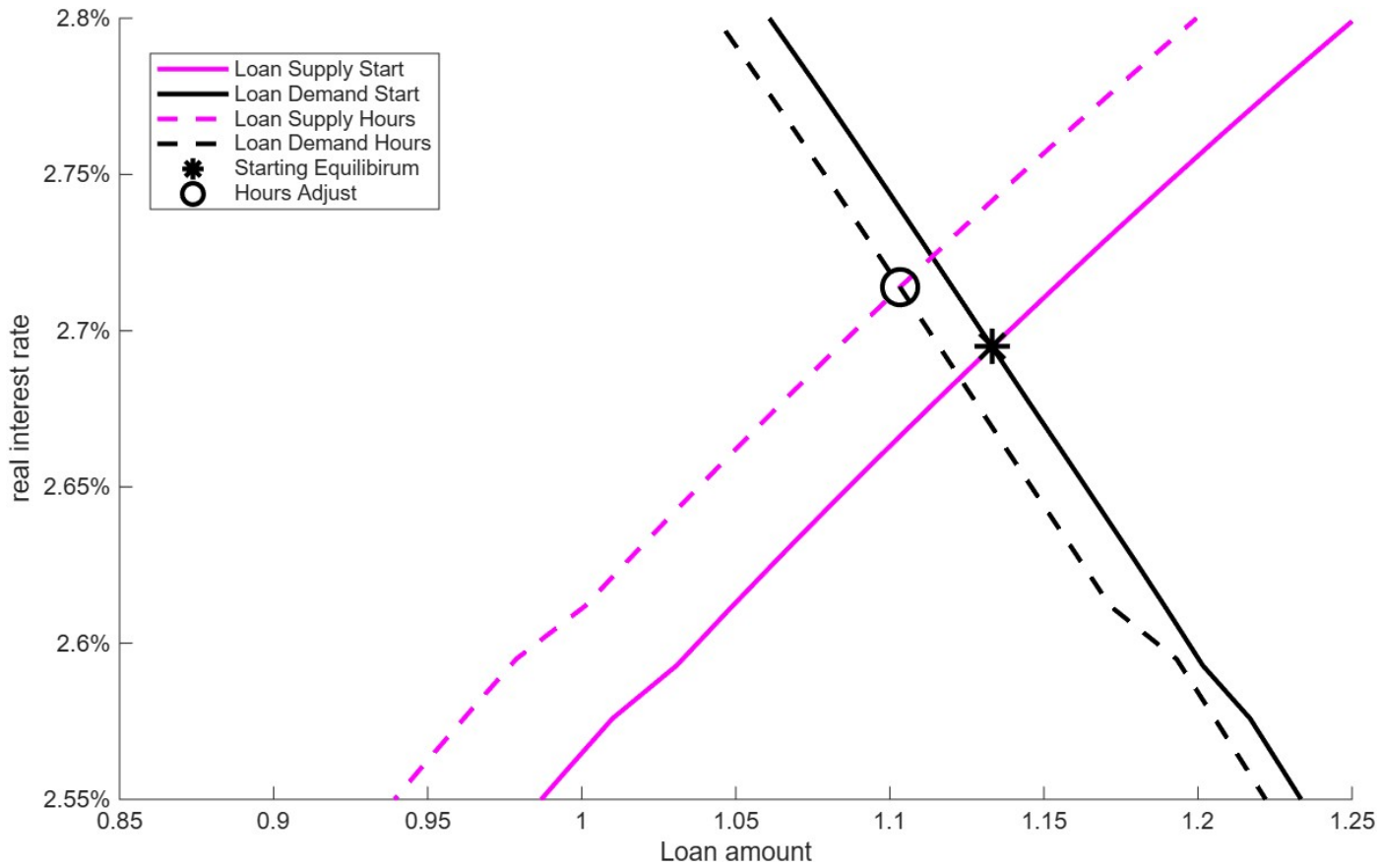
# Results: HA Comparative Static Loanable funds market




*Notes*: Loanable funds market in equilbrium.

Include G and $B_{gov}$

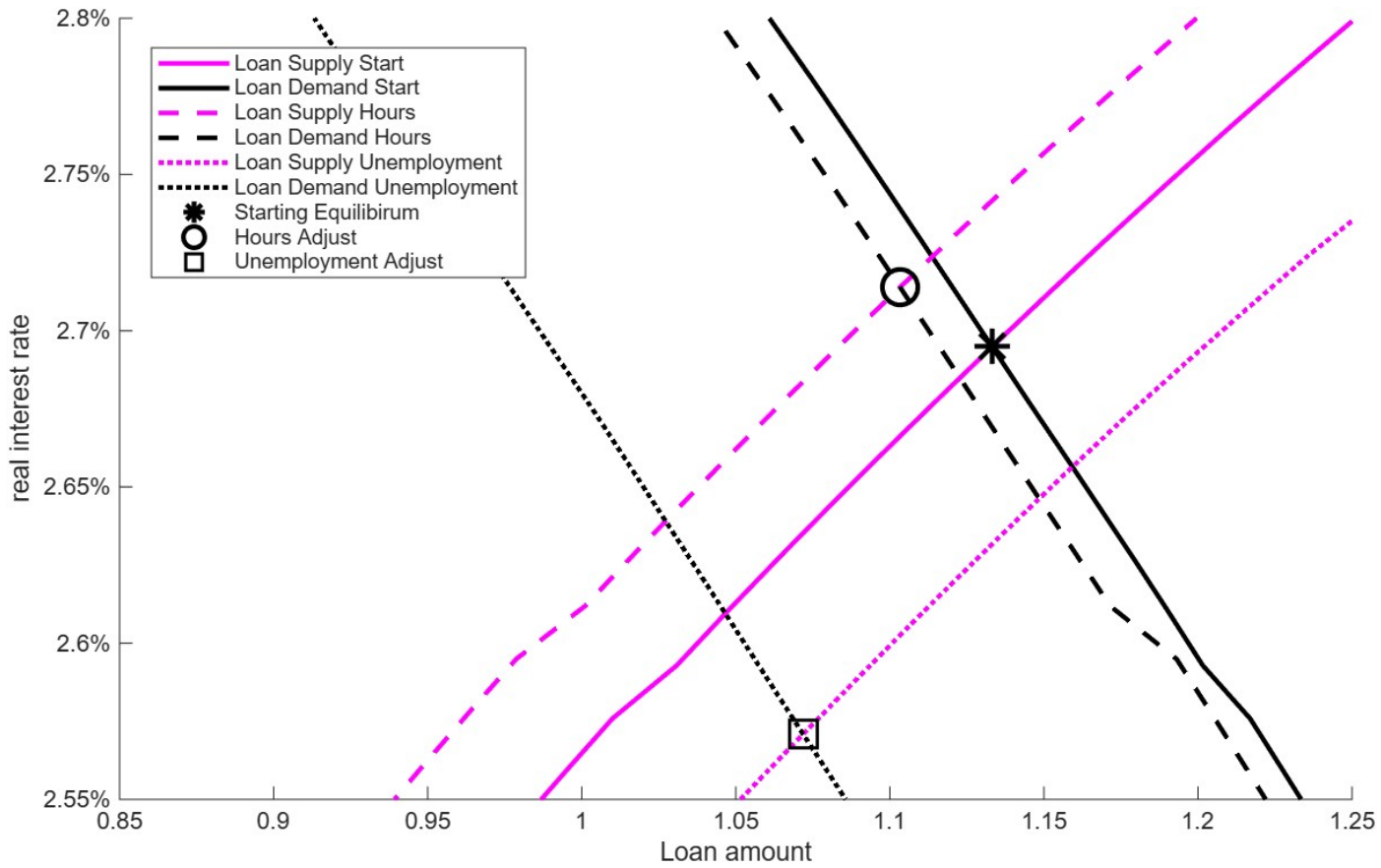
# Results: HA Comparative Static Loanable funds market




*Notes*: Loanable funds market when labor adjusts over the intensive margin after a shock.

Include G and $B_{gov}$

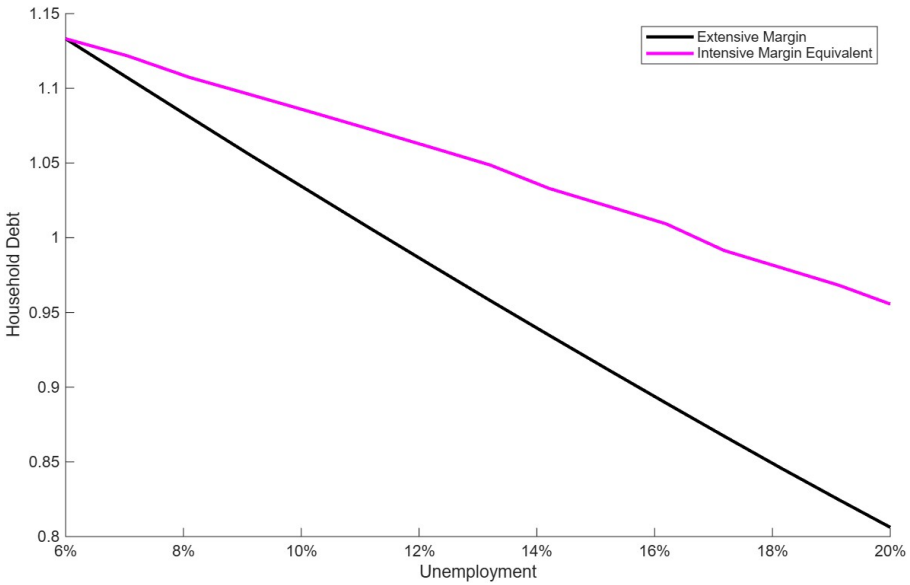
# Results: HA Comparative Static Loanable funds market



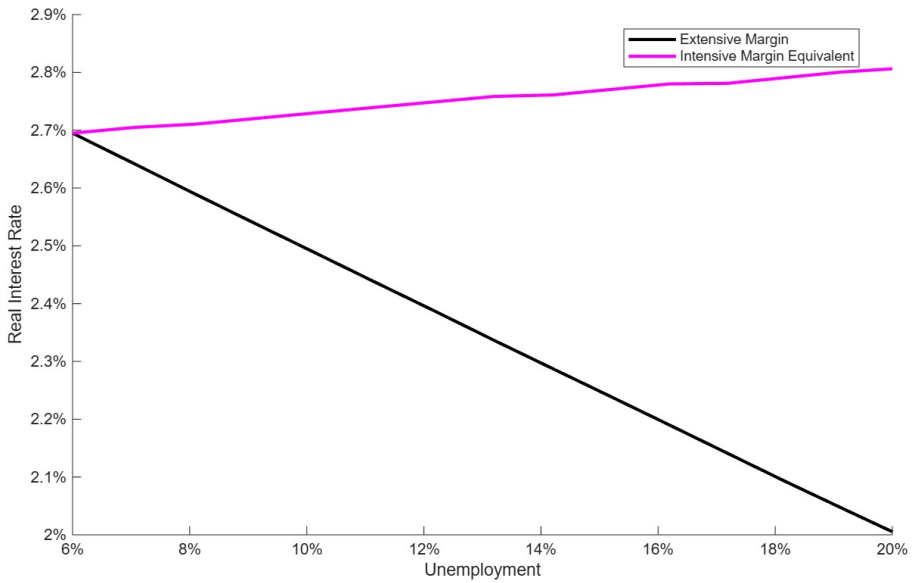

*Notes*: Loanable funds market when labor adjusts over the extensive margin after a shock.

Include G and $B_{gov}$

# Results: Comparative Static Loan Quantity and Interest Rate

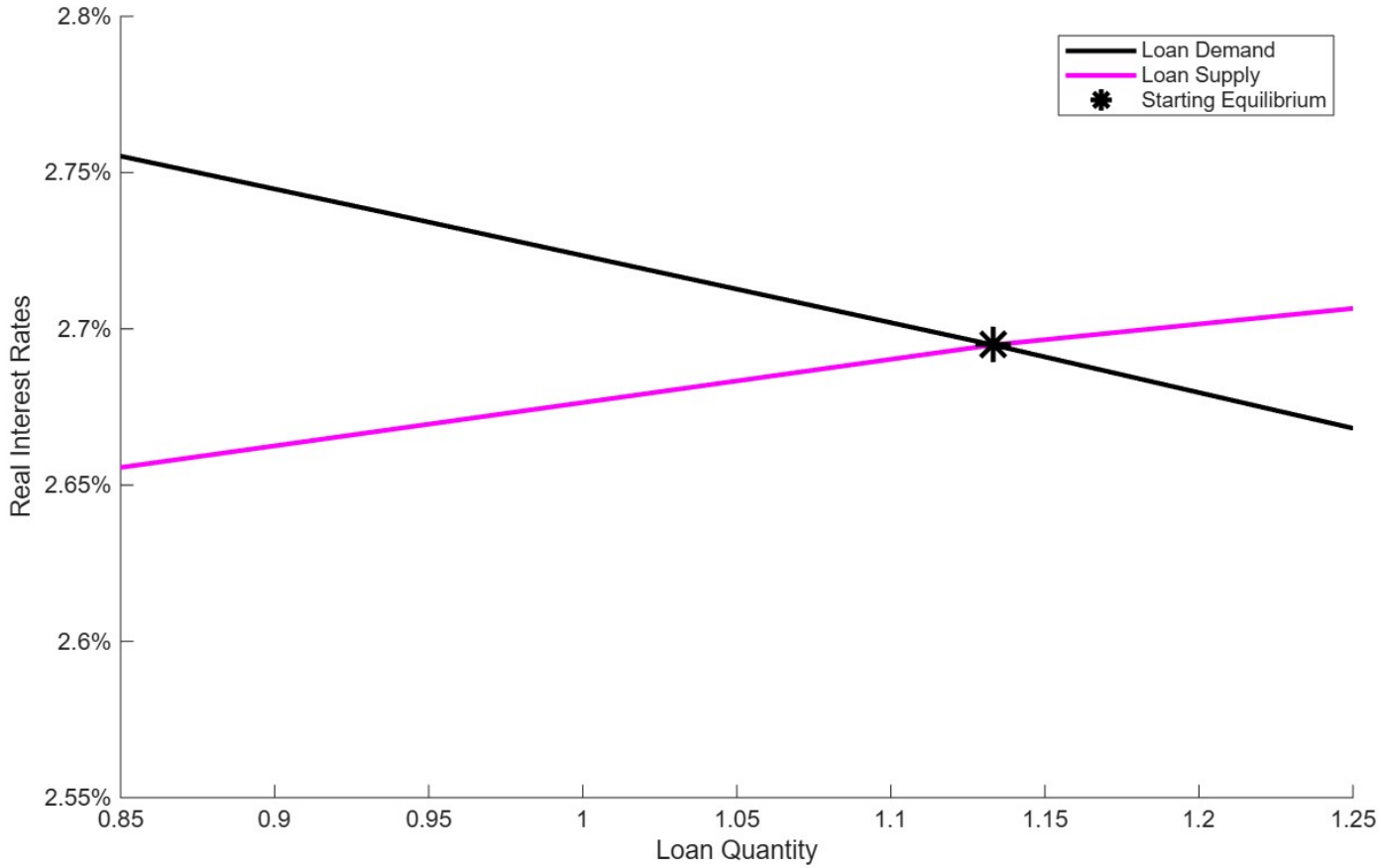



*Notes*: The loan amount drops at faster over the extensive margin.



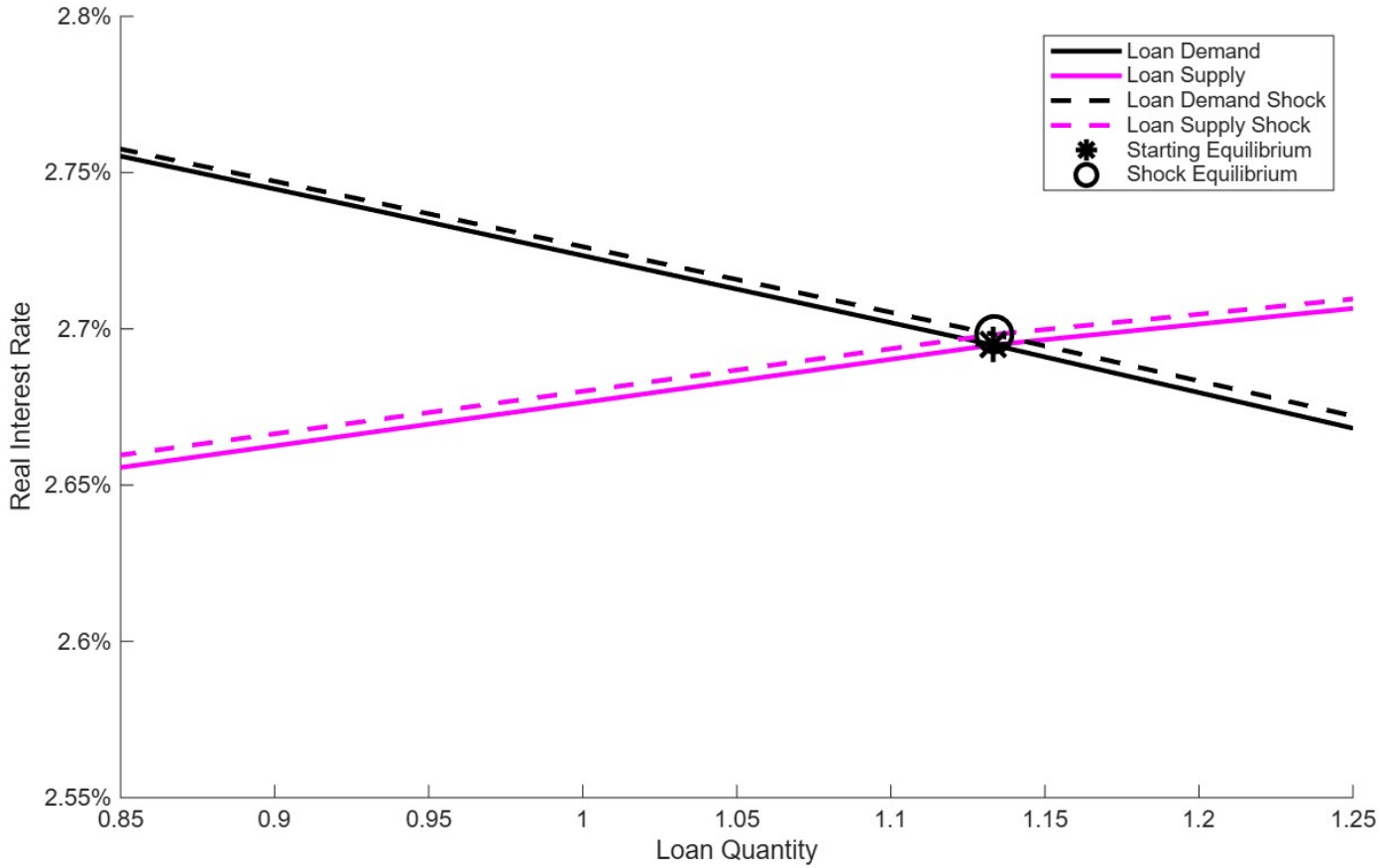

*Notes*: Interest rates drop over the extensive margin and rise in over the intensive margin.

# Results: HA Comparative Static Modified Huggett




*Notes*: The steady state loanable funds market shows much larger elasticity for both supply and demand

# Results: HA Comparative Static Modified Huggett




*Notes*: There is little to no effect in the comparative static for the Huggett model.

# HA IRF Overview

**Benefits of IRF**

- IRF allows us to compare the IRF with both temporary and permanent productivity shocks in the intensive and extensive margin models.
- This will highlight the impact of incorporating employment risk on the saving and borrowing dynamics.

**Variables of Interest:**

- Household loans outstanding is the key variable of interest.
- Real interest rates, percent of individuals borrowing constrained, hiring transition probabilities, and the unemployment rate are recorded as a full dashboard.

# IRF Methodology

## Overall Methodology

- I use the Achdou, Han, Lasry, Lions, and Moll (2022) continuous time approach to solve the model.
- The value function (HJB) is solved backward, and then the distribution is solved forward (KFE).

## Labor Adjustment Cost

- This is added because transitioning employment instantaneously doesn't make sense.
- Lechthaler & Snower (2013) added quadratic labor adjustment costs to a new Keynesian model.

# Solving the IRF

## HJB and KFE

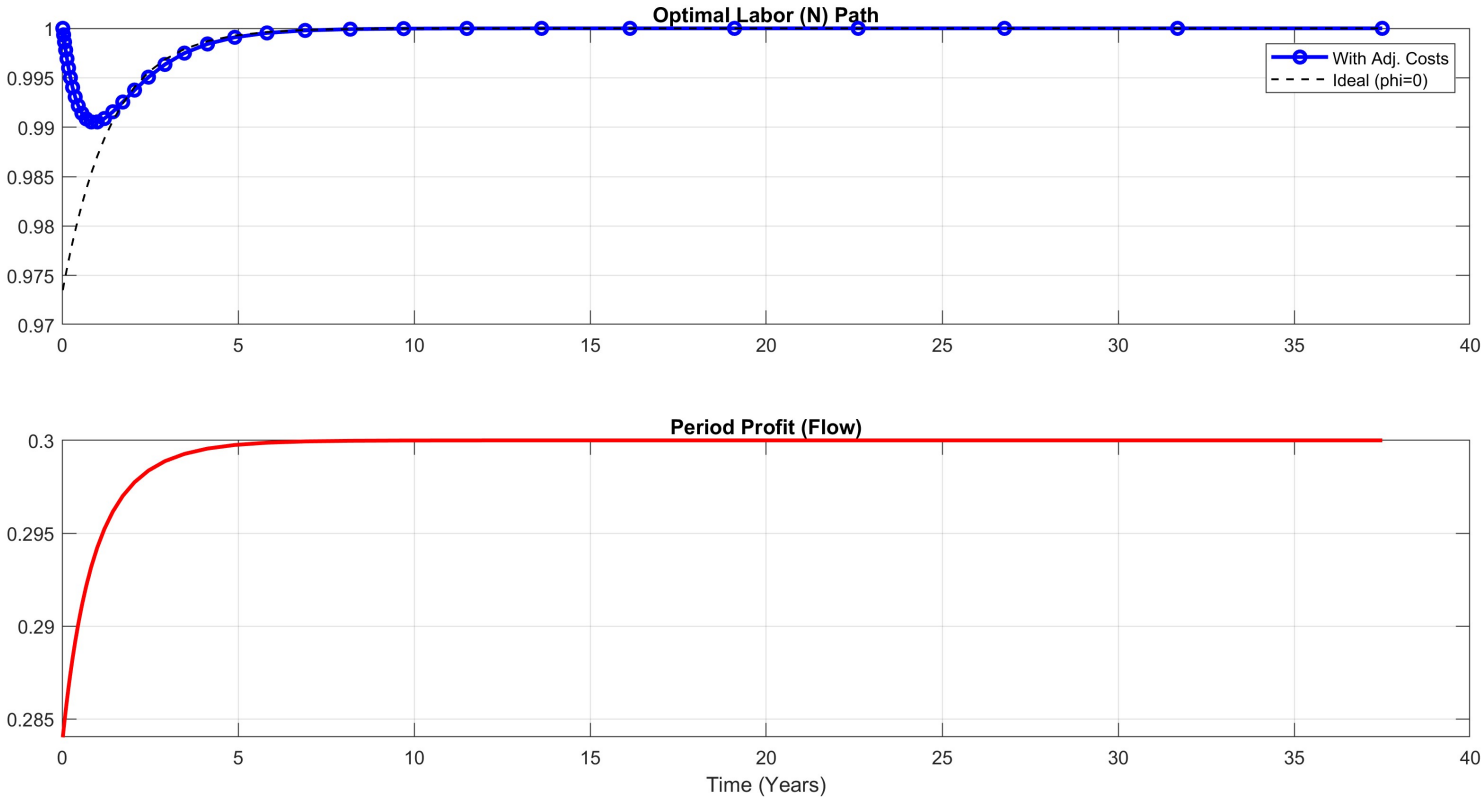
- HJB is solved backward, and KFE is solved forward.
- Three residuals: labor market, government budget, and asset residual are solved in order.

**Effective Labor Determined by:**

$$V = \max_{N_t} \int_0^\infty e^{-\rho t}(A_t(N_t)^{1-\alpha} - wN_t - \frac{\theta}{2}\dot{N_t}^2)\,dt \tag{1}$$

- Labor selected maximizes the firm's profit function.
- Adjust the hiring rate in the Extensive Model.
- Adjust hours worked by agent in the Intensive Model ($N_t$).

# Labor IRF




*Notes*: Adding adjustment costs dampens and delays the labor response.

# Solving the IRF cont.

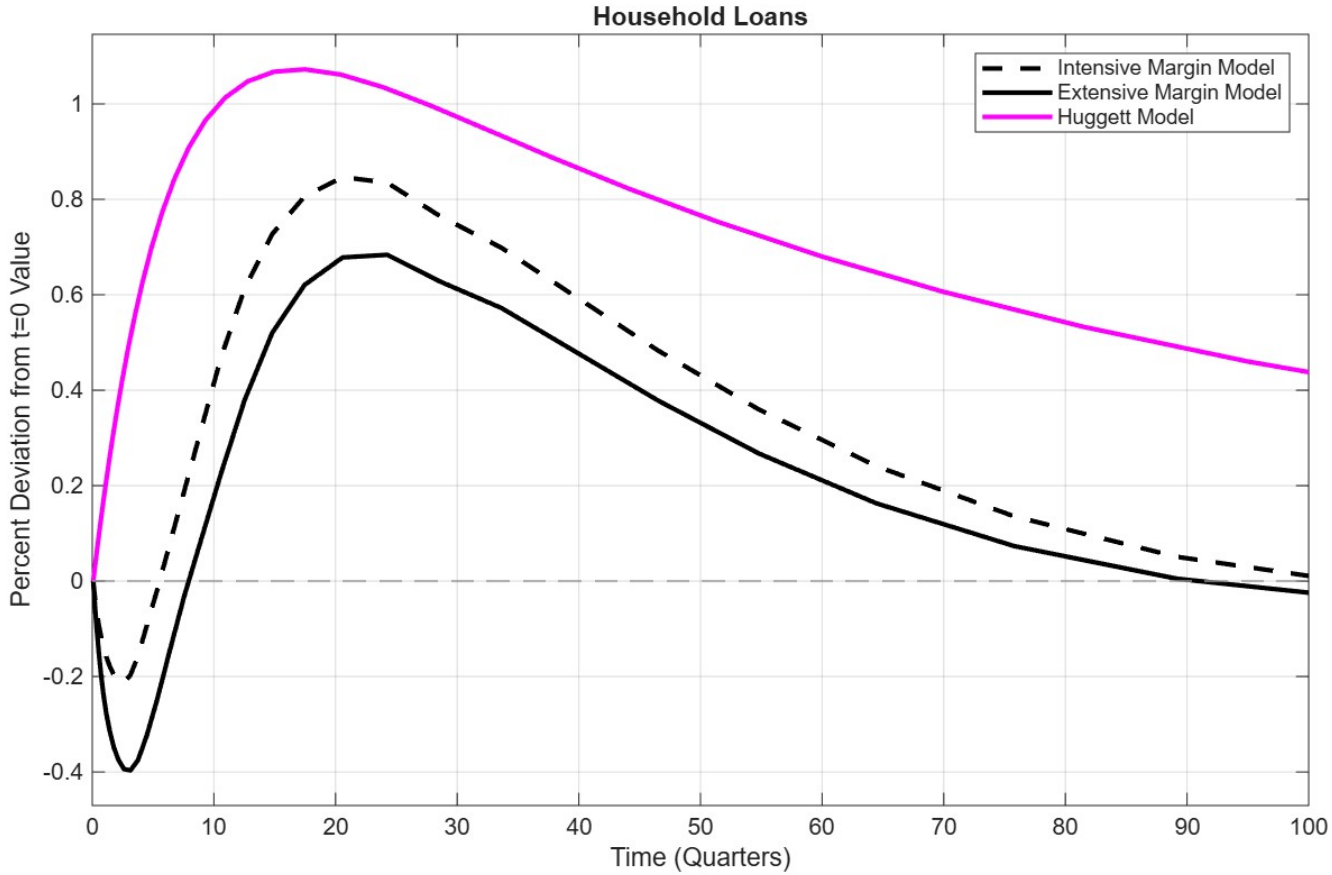
**Taxes**

- Taxes depend on the wage income, dividend income, and the number of unemployed agents. This can be solved directly after the labor market.

**Interest Rate**

- Interest rates have no effect on the labor market and taxes, so they can be solved independently after the labor market and government budget.
- Clear asset market $A(t) = 0$ at all time
- A smoothing function is used to smooth the interest rate path.

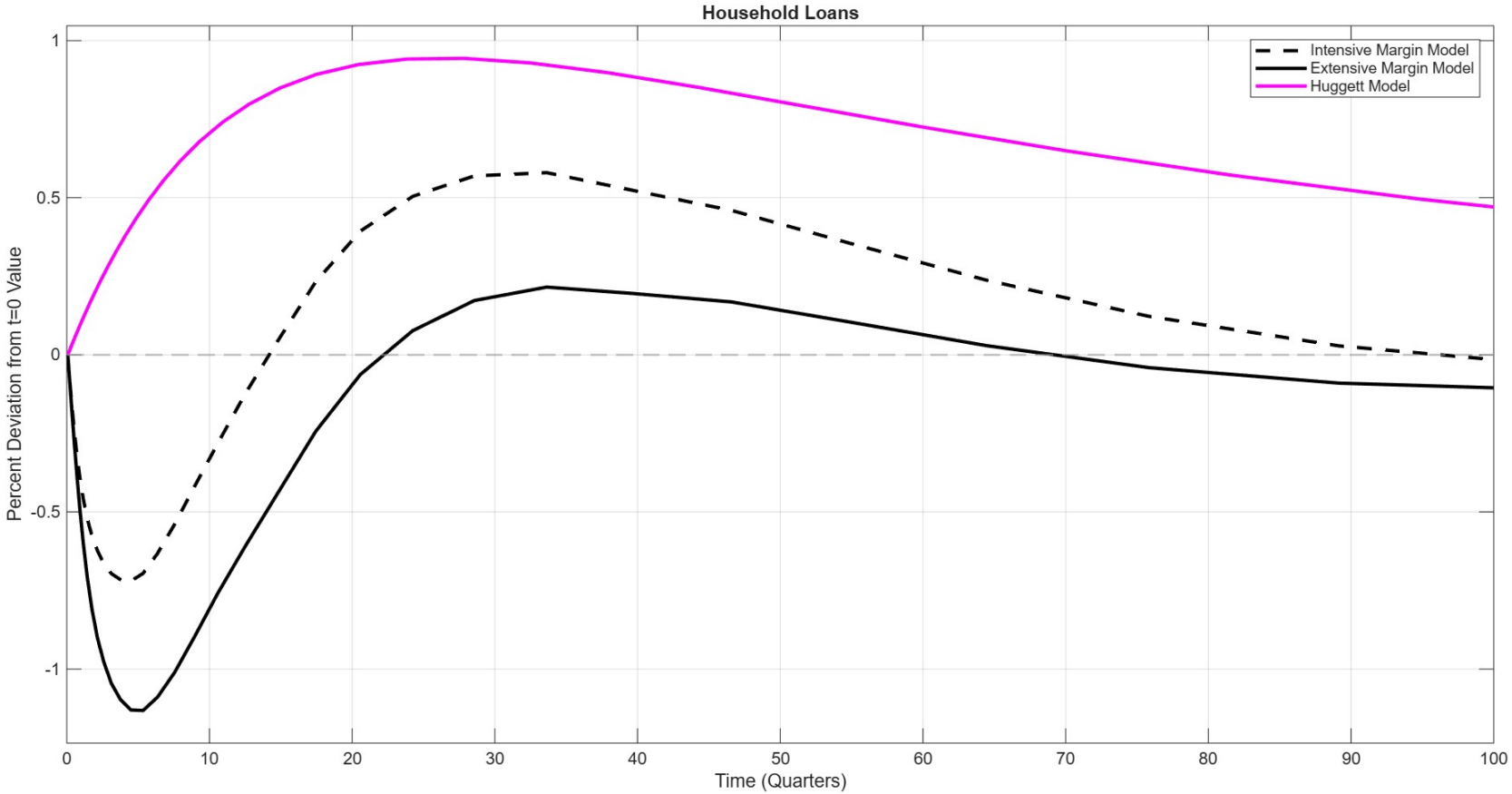
# Temporary Negative Productivity Shock




*Notes*: The extensive margin model does a good job of matching dynamics for 5 quarters before loan quantity overshoots.

Full Dashboard

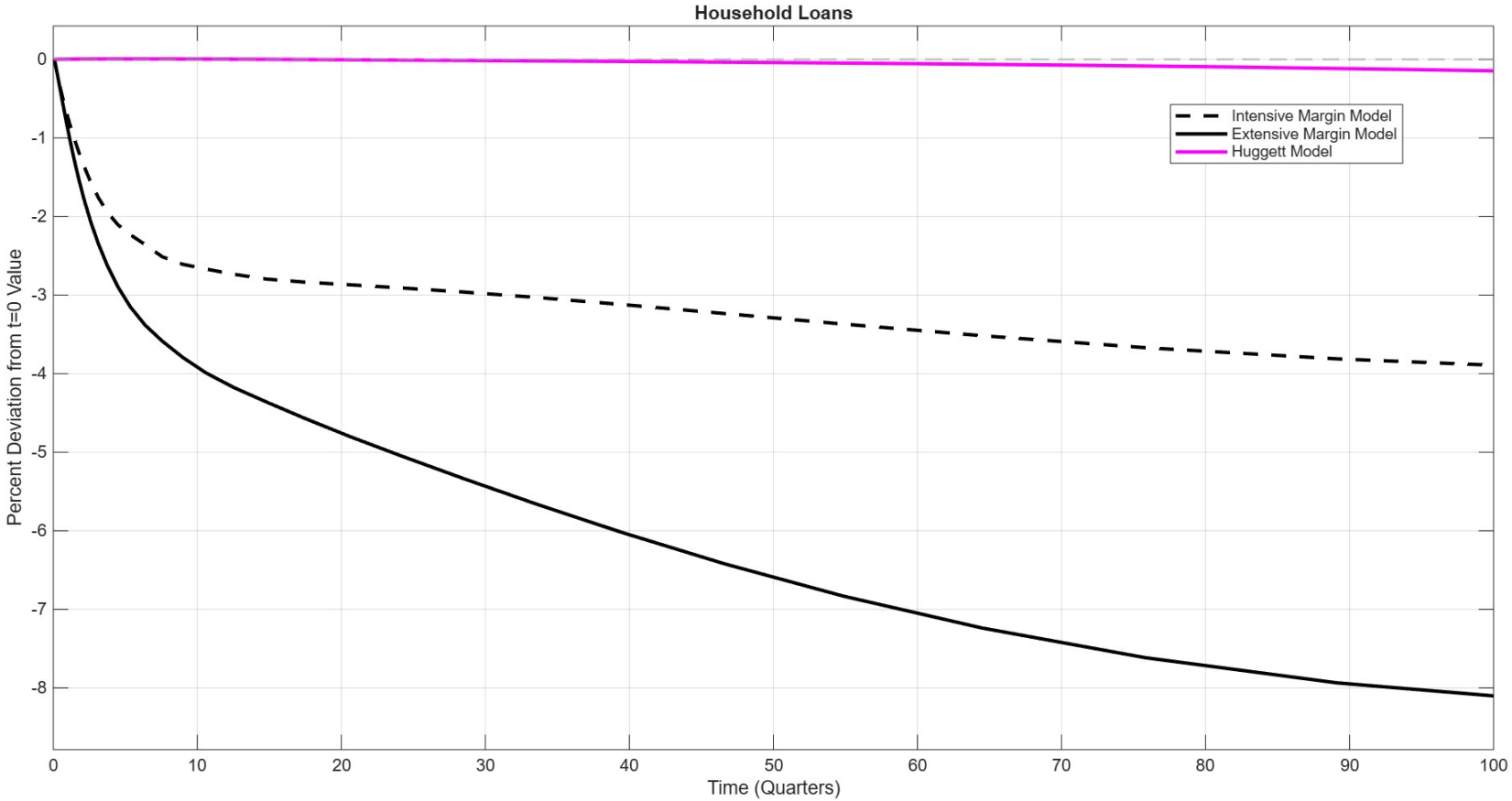
# Persistent Productivity Shock




*Notes*: A more persistent shock deleverages household debt further and reduces overshoot.

Full Dashboard

# Permanent Negative Productivity Shock




*Notes*: Large difference in deleveraging.

Full Dashboard

# Empirical Overview

- Vector autoregression (VAR) and Local Projections (LP) estimate the effect of a productivity shock on real-world data.
- Quarterly variables: labor productivity, weekly hours, unemployment rate, and real household debt.
- Weekly hours and unemployment capture employment dynamics.
- Real household debt is the most stable series to compare with the theoretical models, because interest rates are heavily affected by default probabilities.

# Data used

**US Data:**

- Data is Quarterly from 1952 to 2019
- US lacks state level household debt data.

**European Data As a Complement:**

- Data is Quarterly from 1996 to 2019
- Can be used to split countries into high and low characteristics

# VAR Structure

- The VAR was calculated with a Cholesky decomposition of errors.
- The order of the Upper triangle was as follows: labor productivity, weekly hours, unemployment rate, and household debt.
- The VAR imposes an AR(1) on the productivity shock, and all other variables are lagged by 4 quarters.

# VAR Formula

**VAR formula:**

$$(I - \Phi_1 L - \Phi_2 L^2 - \Phi_3 L^3 - \Phi_4 L^4) X_t = e_t \tag{2}$$

This is a reduced from VAR that forces labor productivity to be an AR(1)

$\Phi_1$ layout $\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$ $\Phi_{2,3,4}$ layout $\begin{bmatrix} 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$

$$X_t = \begin{bmatrix} A_t : LaborProductivity \\ H_t : WeeklyHoursWorked \\ U_t : UnemploymentRate \\ B_t : RealHouseholdDebt \end{bmatrix}$$

# VAR Error Terms

**Cholesky Error Terms:**

$$e_t = Bu_t$$

**Matrix Form:**

$$\begin{bmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \\ e_{4t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ b_{21} & 1 & 0 & 0 \\ b_{31} & b_{32} & 1 & 0 \\ b_{41} & b_{42} & b_{43} & 1 \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \\ u_{4t} \end{bmatrix}$$

# Local Projection

**Local Projection:**

- Impulse response functions by running a sequence of OLS regressions for different horizons (h).

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma_h x_t + v_{t+h}$$

$\alpha_h$ constant, $\beta_h$ Reaction to shock, and $\gamma_h x_t$ control variables

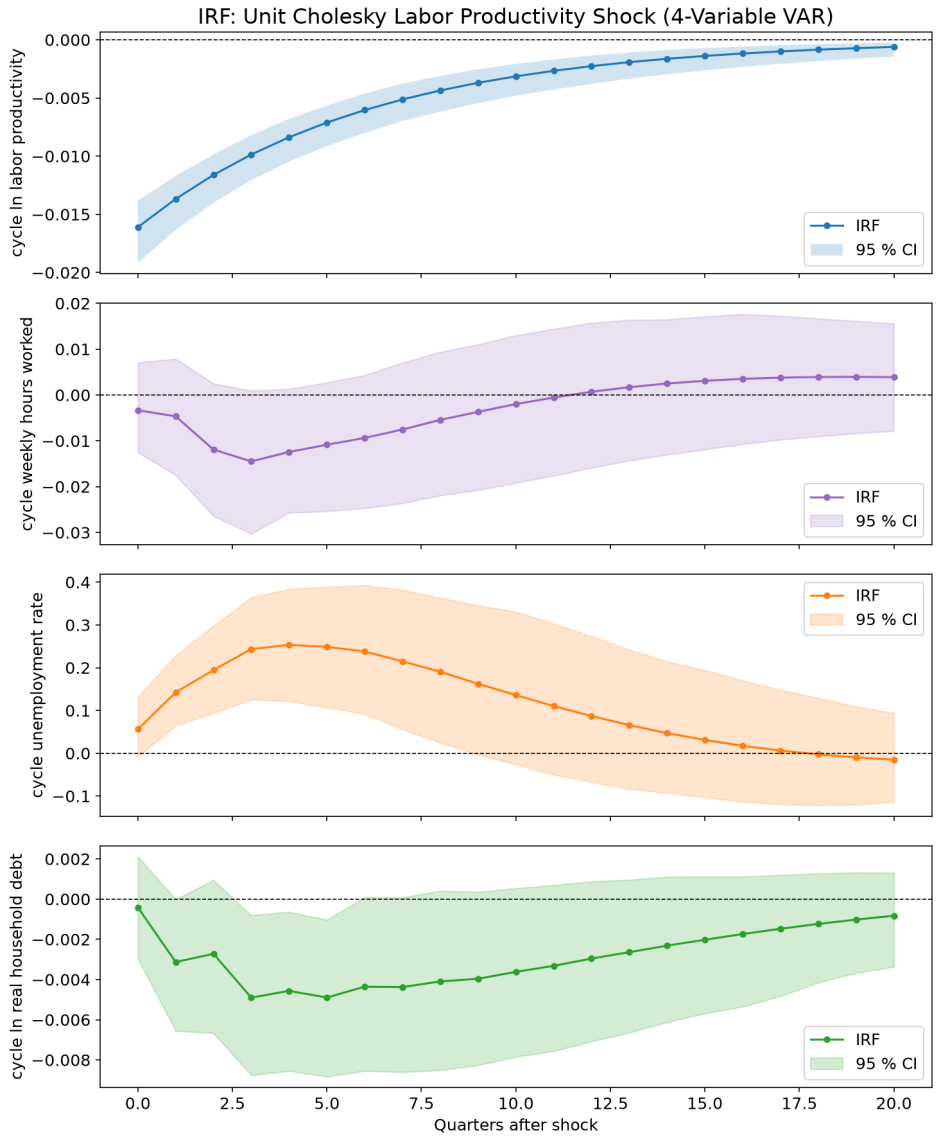
- Residual MA(h) Error Term:

$$v_{t+h} = u_{t+h} + \psi_1 u_{t+h1} + ... + \psi_h ut$$

# VAR European Average

VAR shows the following:

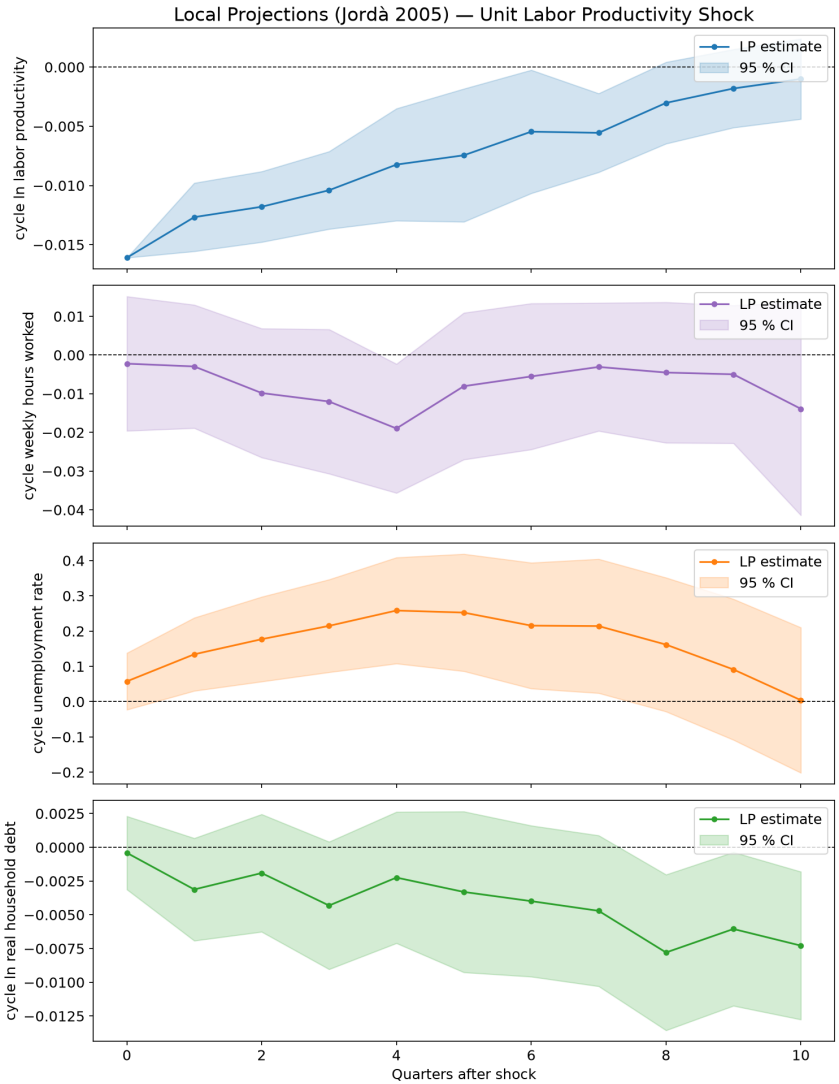
- Ln detrended productivity follows an AR(1) structure and shows the difference in log levels.
- Weekly hours worked don't move significantly.
- The UR increases significantly.
- Ln detrended real debt decreases significantly.



# LP European Average

LP broadly follows the VAR findings:

- Ln detrended productivity now moves freely.
- Weekly hours has minimal movement.
- The UR increases significantly.
- Ln detrended real debt decreases significantly over time.

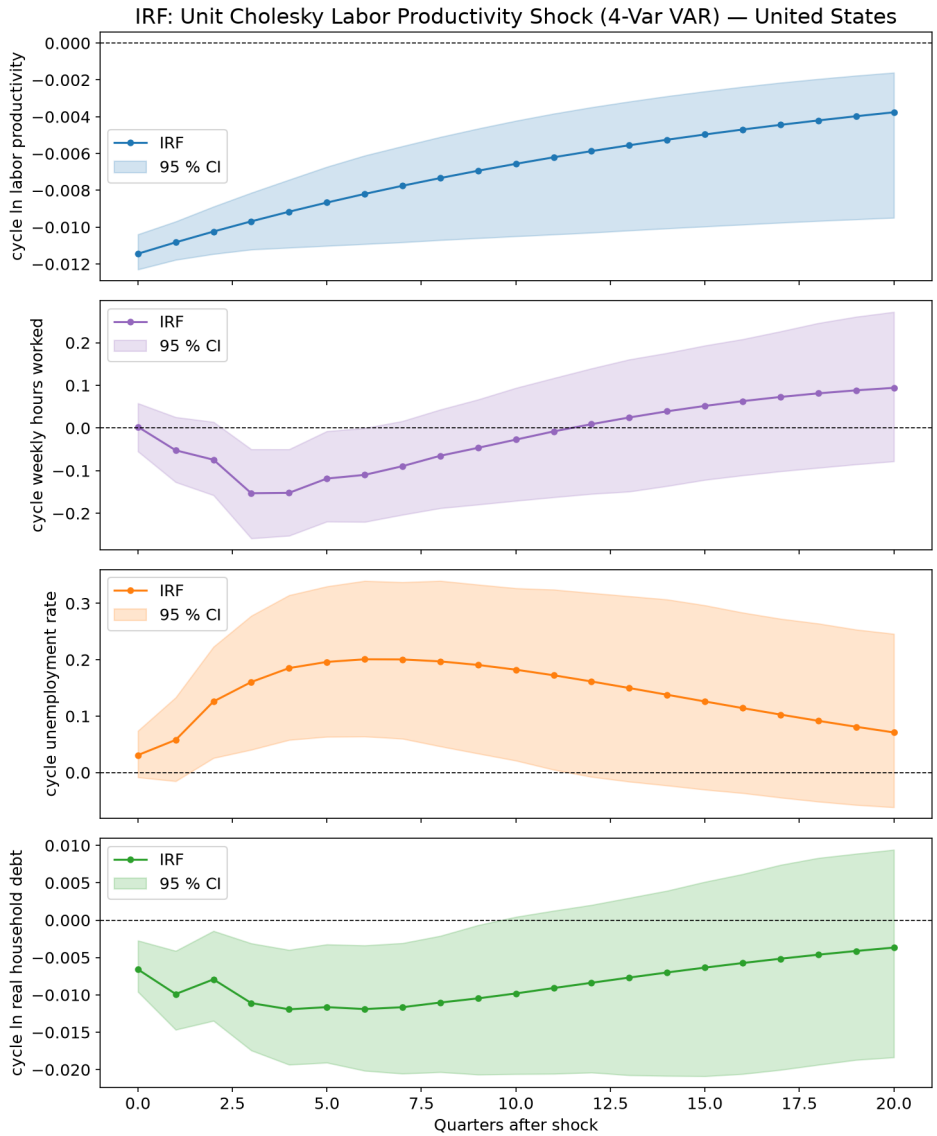



*Notes*: The VAR highlights the same direction of

# VAR US

VAR shows the following:

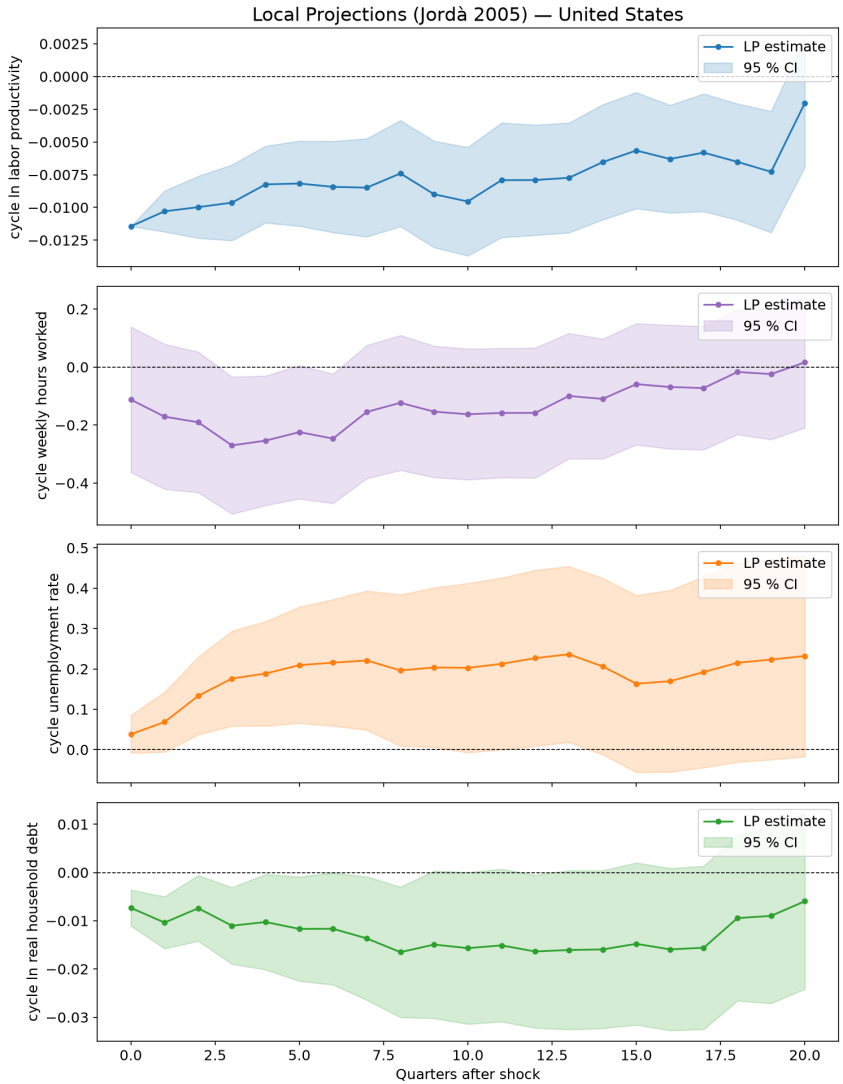
- Ln detrended productivity follows an AR(1) structure and shows the difference in log levels.
- Weekly hours worked don't move significantly.
- The UR increases significantly.
- Ln detrended real debt decreases significantly.



# LP US

LP broadly follows the VAR findings:

- Ln detrended productivity now moves freely.
- Weekly hours has minimal movement.
- The UR increases significantly.
- Ln detrended real debt decreases significantly over time.




*Notes*: The VAR highlights the same direction of

# Split Countries to Look at Difference

**Why countries are split**

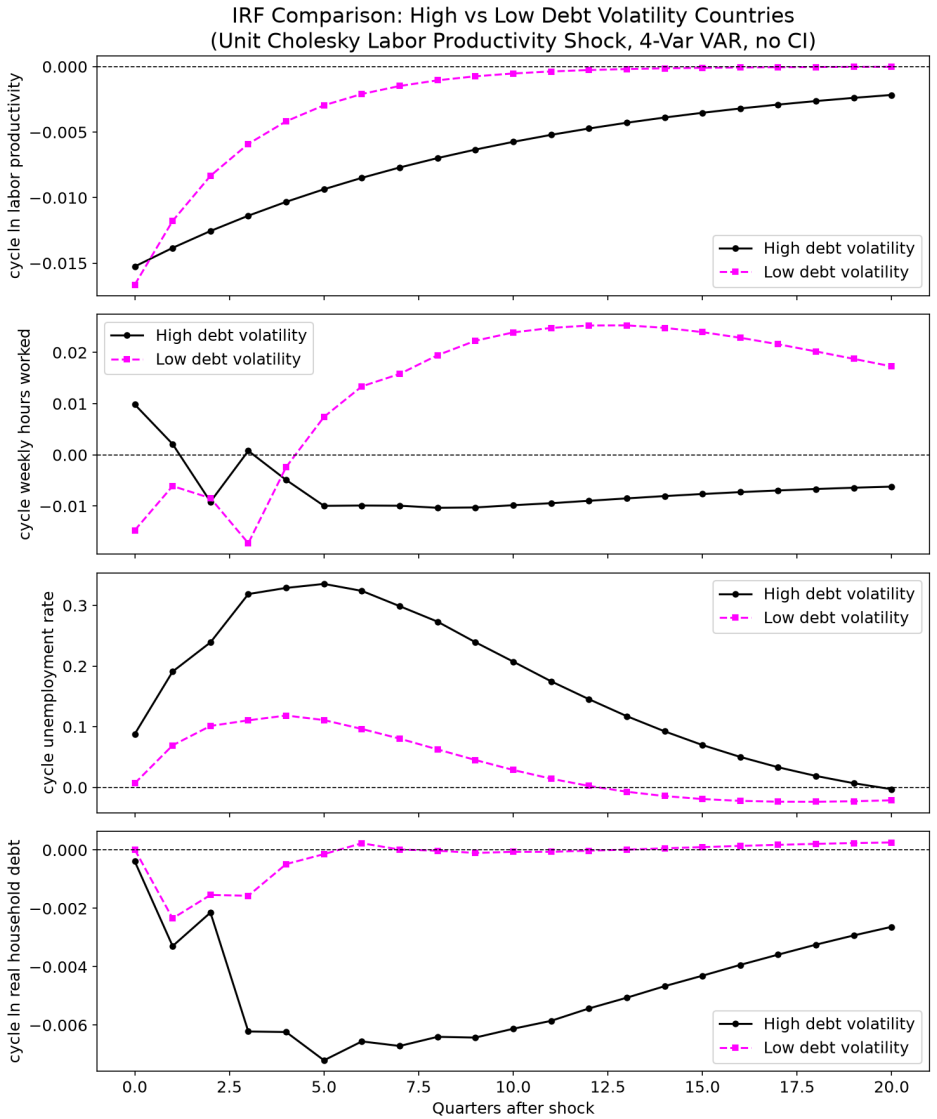
- Countries are split to see if there are significant differences in the IRF based on the splits

**Countries Split by the following Groups**

- Household Debt Variance
- Household Debt to GDP

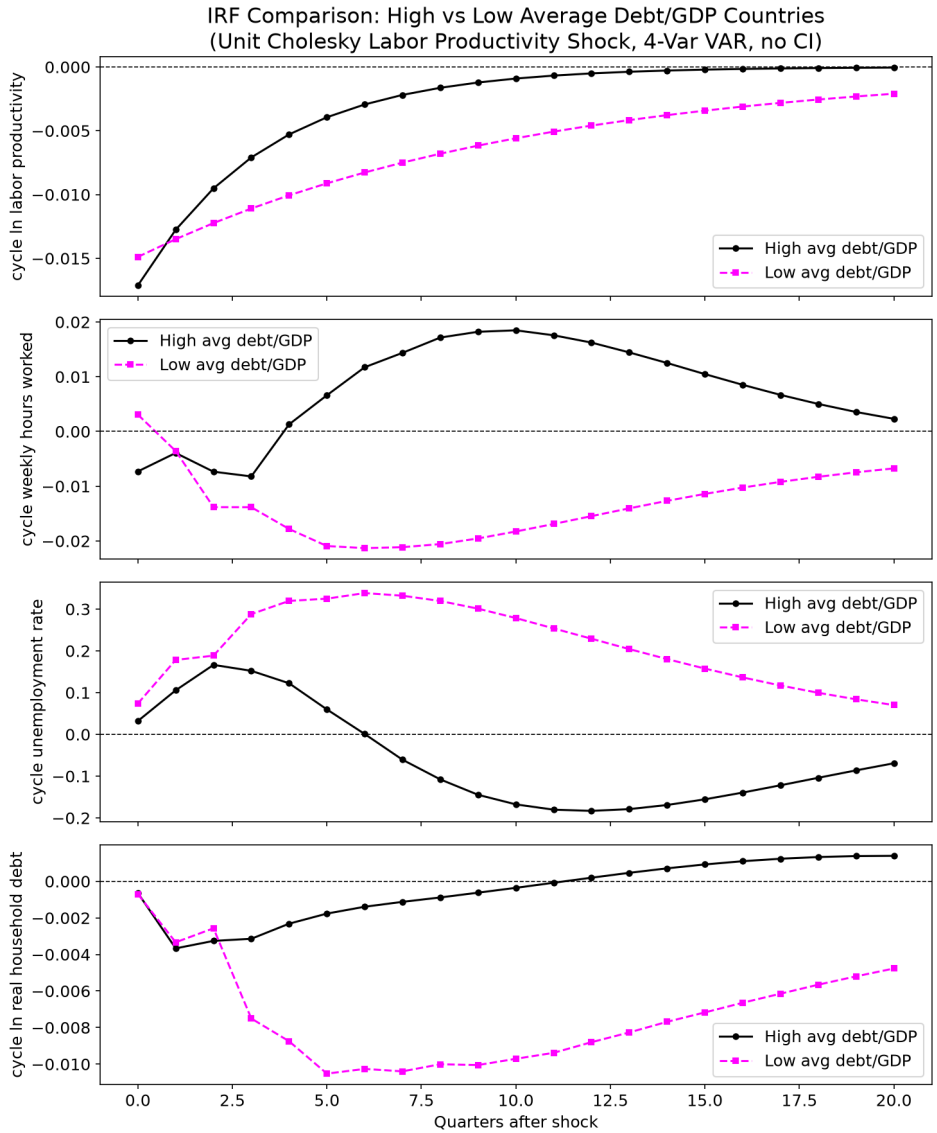
# VAR Household Debt Volatility

- The countries are split based on the variance in cyclical real debt.
- The median variance is 0.003472.
- High debt volatility countries (8): Czechia, Finland, Greece, Hungary, Italy, Poland, Portugal, and Spain.
- Low debt volatility countries (8): Austria, Belgium, Denmark, France, Germany, Ireland, Norway, and Sweden.



# VAR High vs Low Household Debt/GDP

- The countries are split by on mean debt to GDP ratios.
- Median loans-to-GDP: 51.0
- High average household debt/GDP countries (8): Denmark, France, Germany, Ireland, Norway, Portugal, Spain, and Sweden.
- Low average debt/GDP countries (8): Austria, Belgium, Czechia, Finland, Greece, Hungary, Italy, and Poland.



# Empirical Findings

- The empirical responses to productivity shocks show that the movement in the unemployment rate is significant, and weekly hours are not. They also show that real household loans are significantly affected by productivity shocks.
- Countries with high cyclical real debt variation and lower average household debt to GDP have similar productivity shocks, but the persistence is longer.
- This implies that countries experiencing persistent shocks experience larger increases in unemployment and, therefore, larger decreases in household loan balances.

# Conclusion

- This model captures the "double shock" of unemployment in a life cycle model designed to assess its implications.
- The extensive margin model highlights a drop in real interest rates, and loan quantities following a negative productivity shock.
- These findings are consistant with the empirical findings that the VAR and LP generated.

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# Lack of ability to spend and borrow from external sources

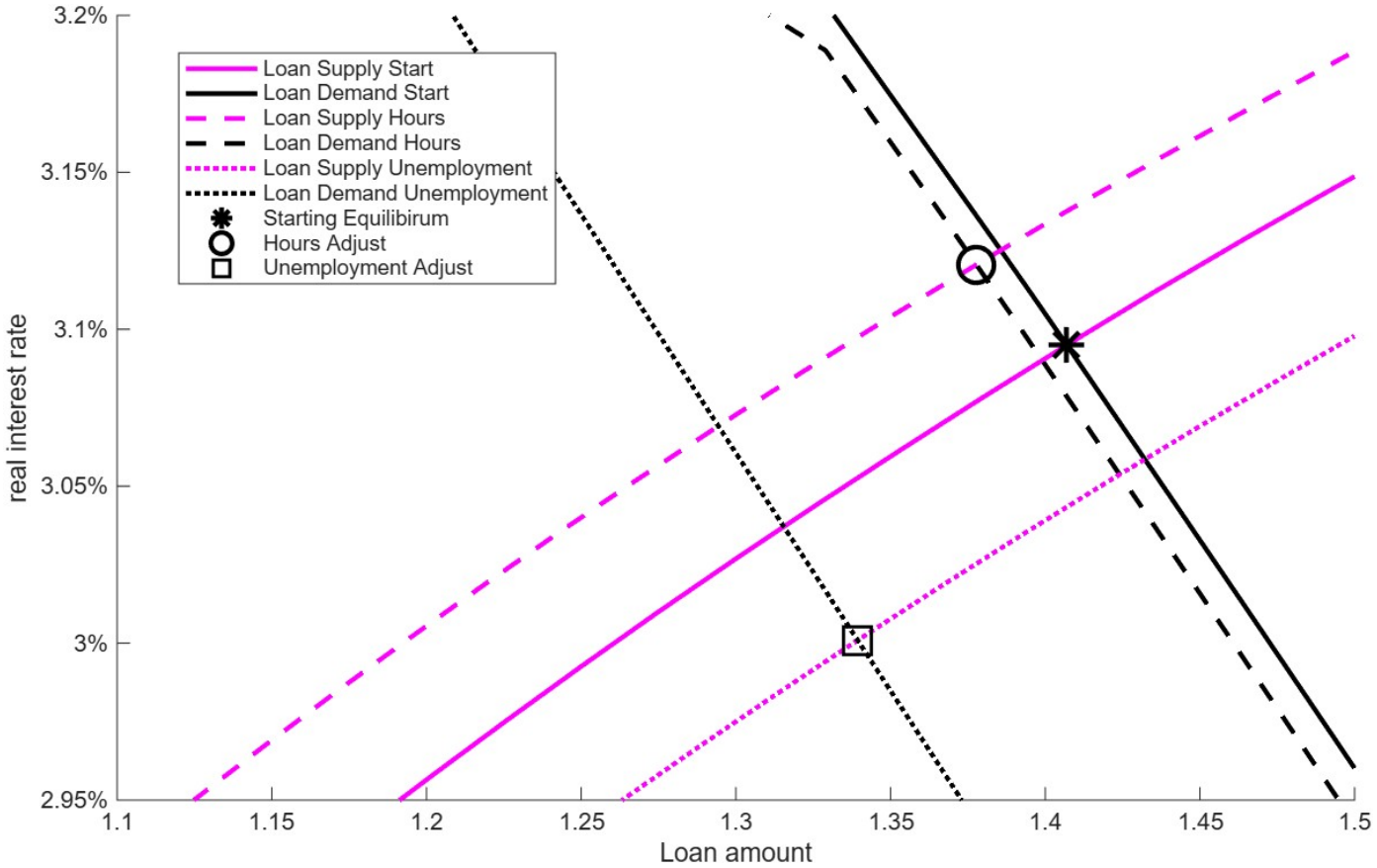
**Forbes Article: 63% Of Americans Don't Have Enough Savings To Cover A $500 Emergency**

- This model simplifies all assets into one account, where in reality, individuals might have loans and a positive savings account.
- This is limited because most people don't have that much money in an extra savings account.

**Liberty Street Economics: 18 percent of borrowers were using at least 90 percent of their available credit**

- Credit cards allow people to smooth consumption even if they are in debt.
- This is limited because a good number of individuals are already at or near their credit card limit.

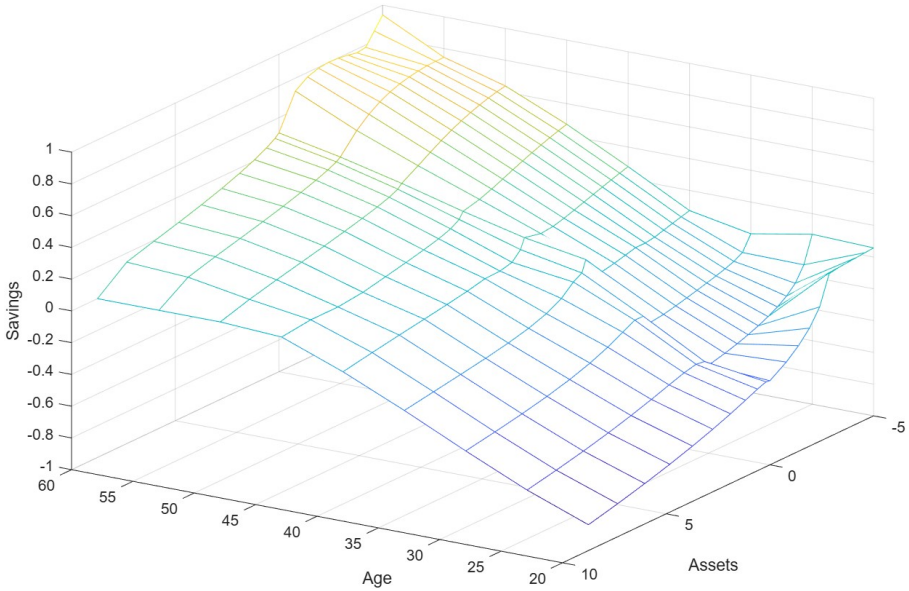
# HA Comparative Static with G and $B_{gov}$



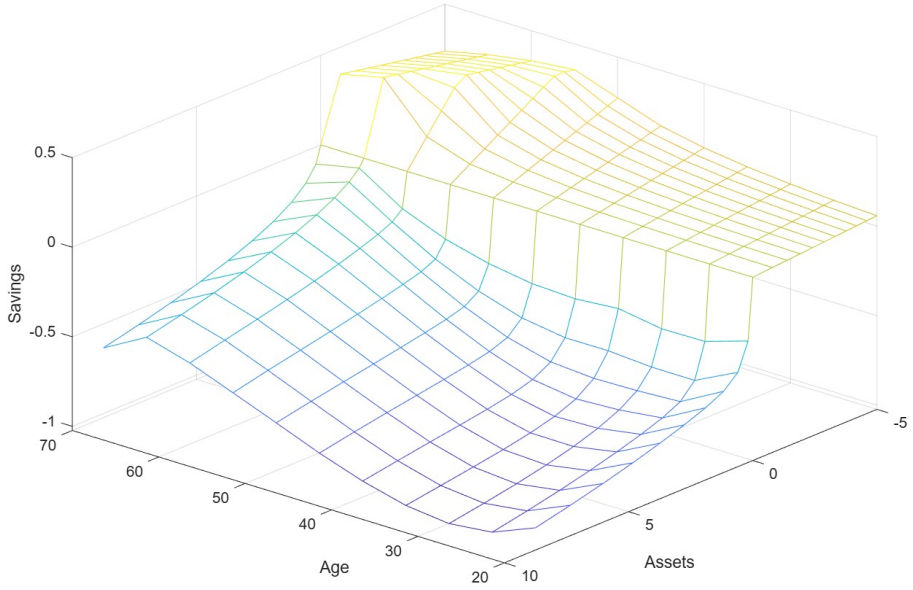

*Notes*: With G is set to .17 and $B_{gov}$ is set to 70% annual GDP, the movement in supply and demand are reiterated

Return

# HA Saving Curves

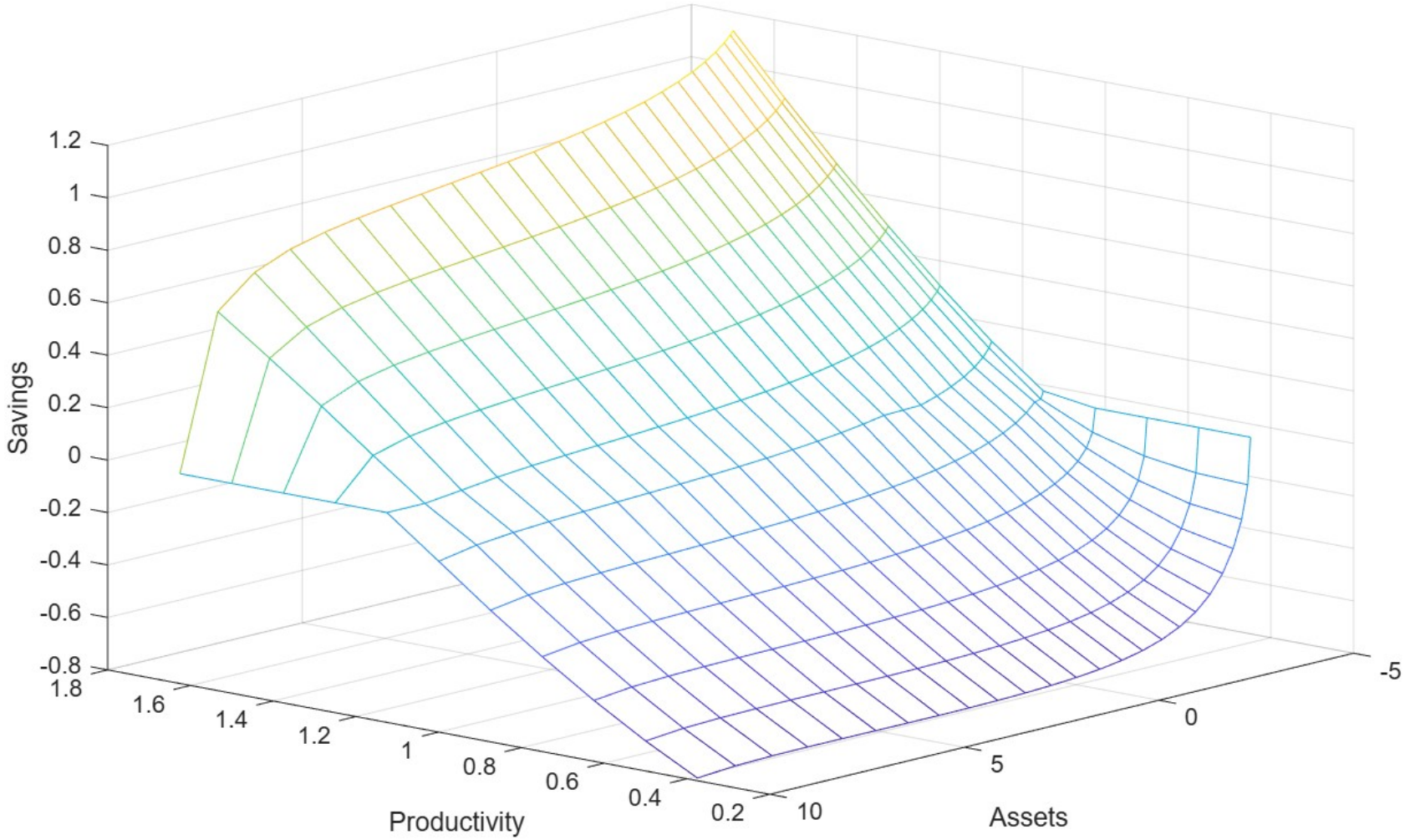



*Notes*: Saving is higher with age and with lower asset levels.




*Notes*: Saving is negative above the zero asset borrowing constraint.

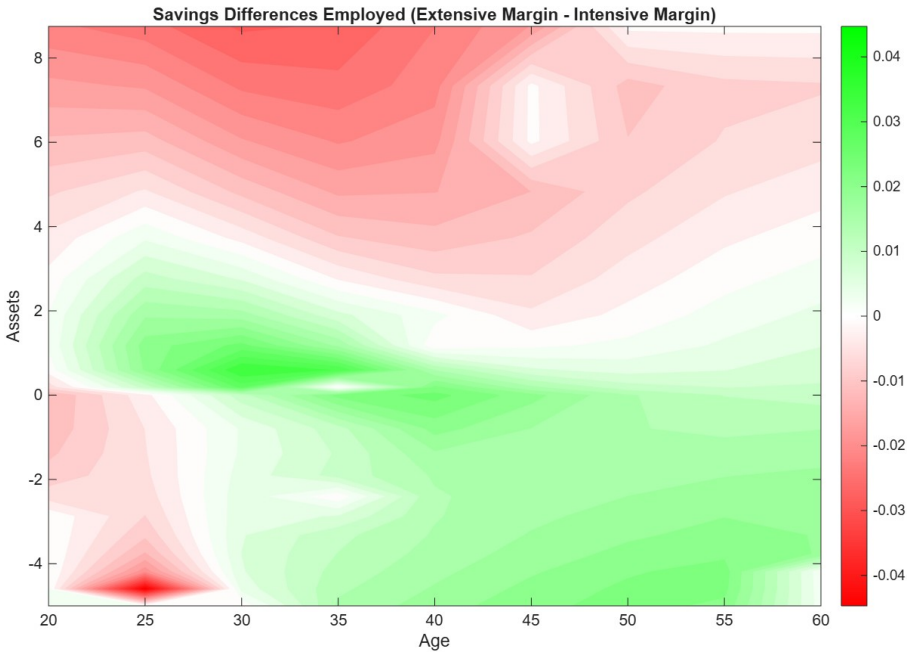
# Modified Huggett Saving Curve



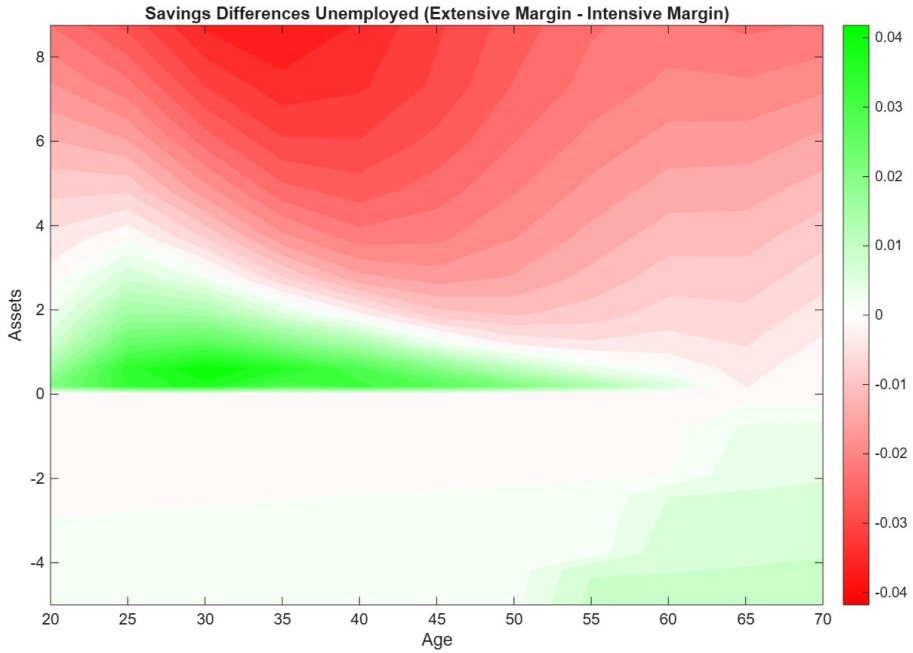

*Notes*:

Much less intricacies than HA model.

# Savings Differential Across the Margins

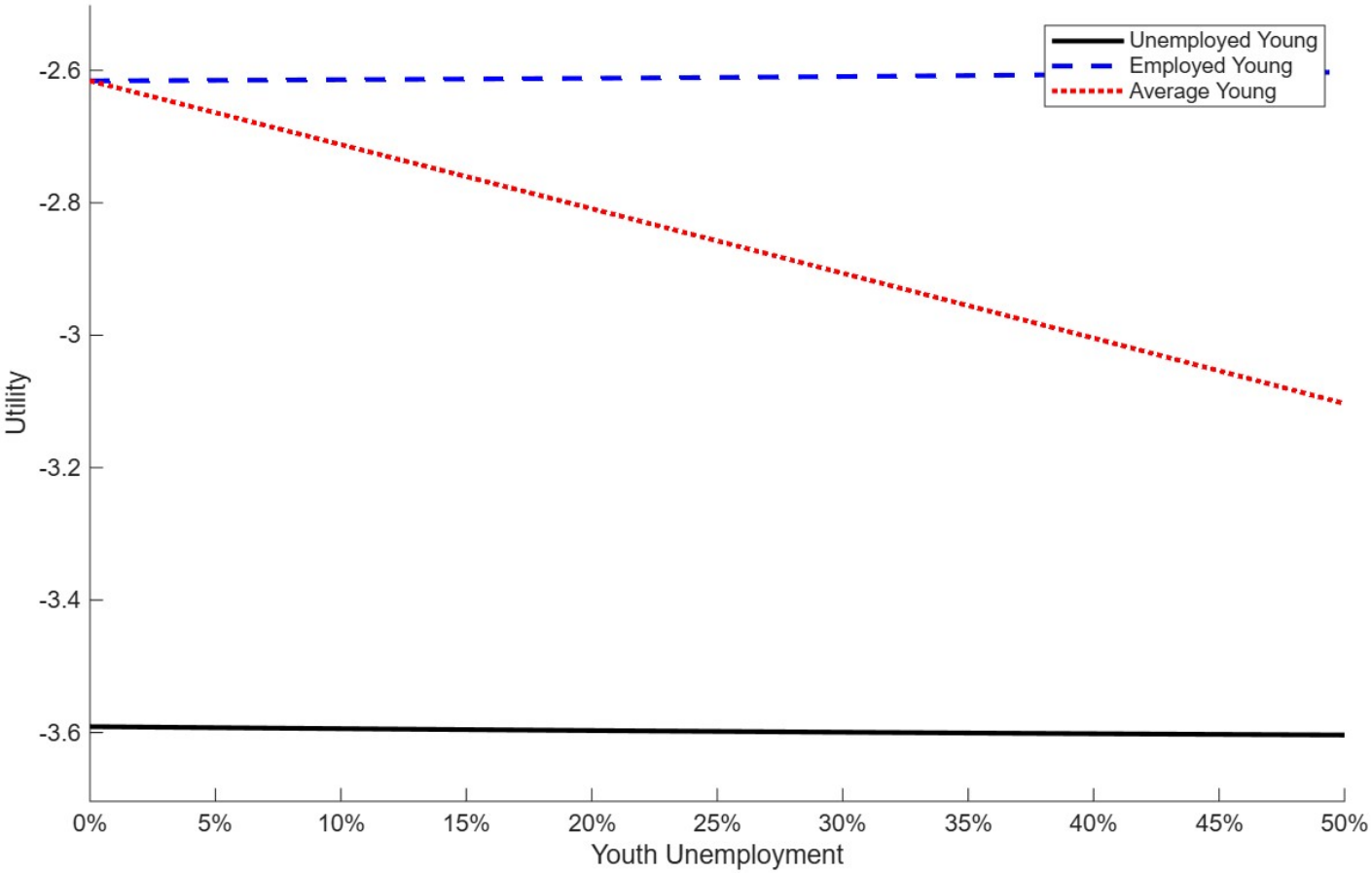



*Notes*: Borrowing is higher for the extensive margin early on, then less as they age.




*Notes*: Precausionary savings increases above zero assets.

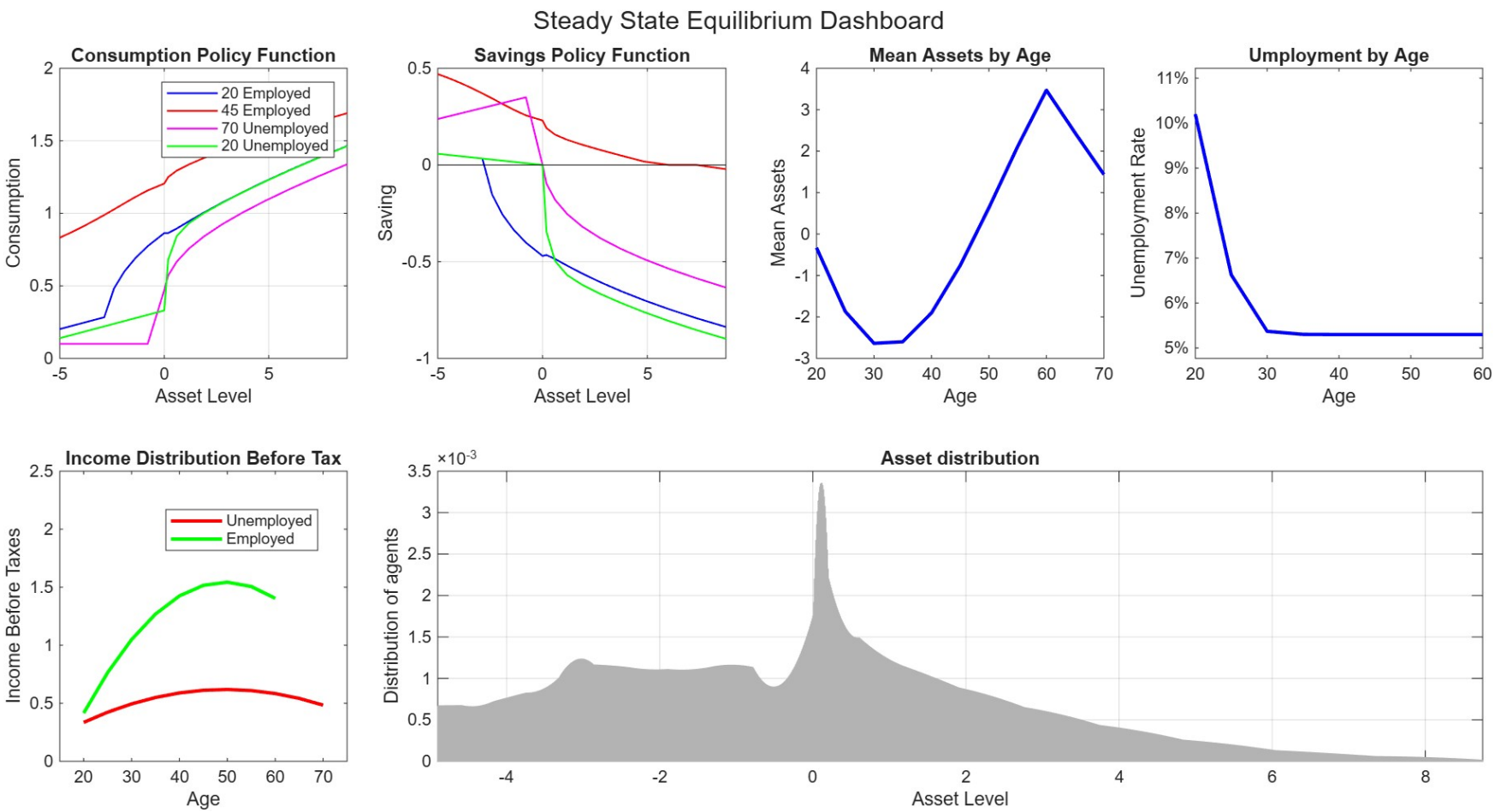
# Results: Youth Utility




*Notes*:

Average youth utility drops when youth unemployment increases.

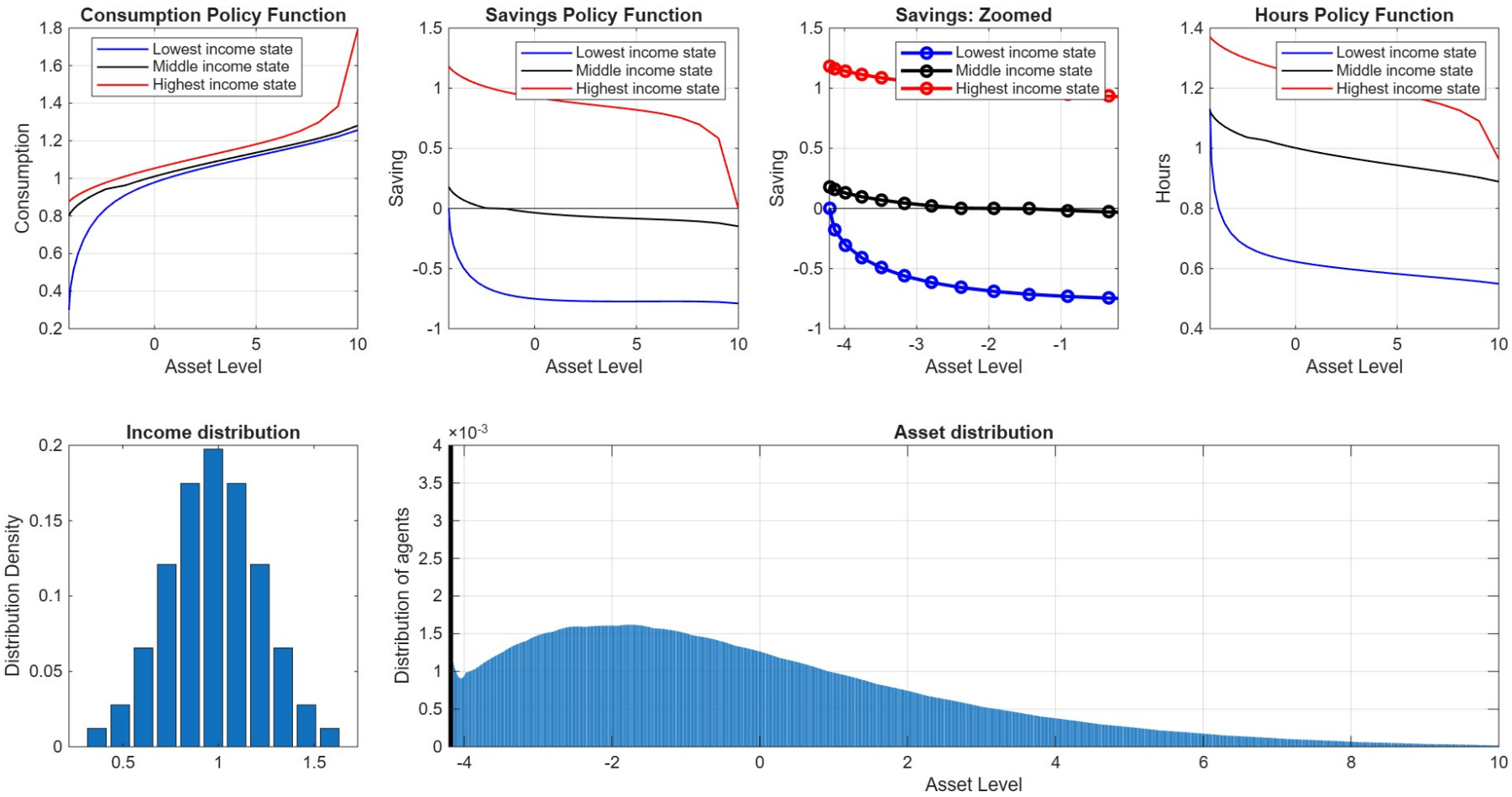
# Steady State Dashboard




*Notes*:

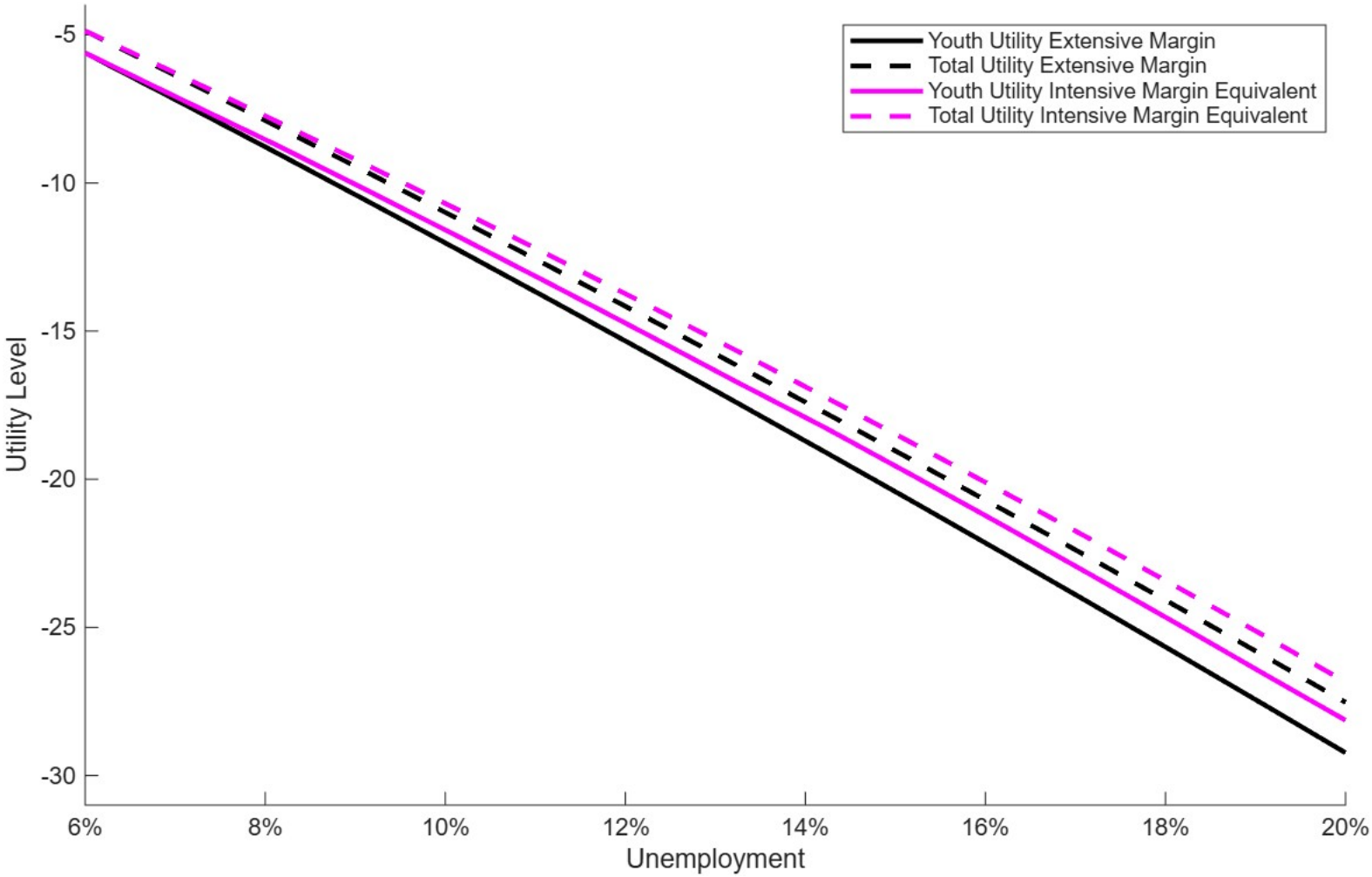
There is no missing middle. Mean assets by age show borrowing and savings behavior because realistic modeling of income and unemployment by age

# Steady State Dashboard: Modified Huggett Model




*Notes*: Asset distribution suffers from the missing middle. Agents here choose how much they work.

# Results: Utility HA model




*Notes*: The extensive margin has a larger utility drop

# Introducing Countercyclical Transfer Payments

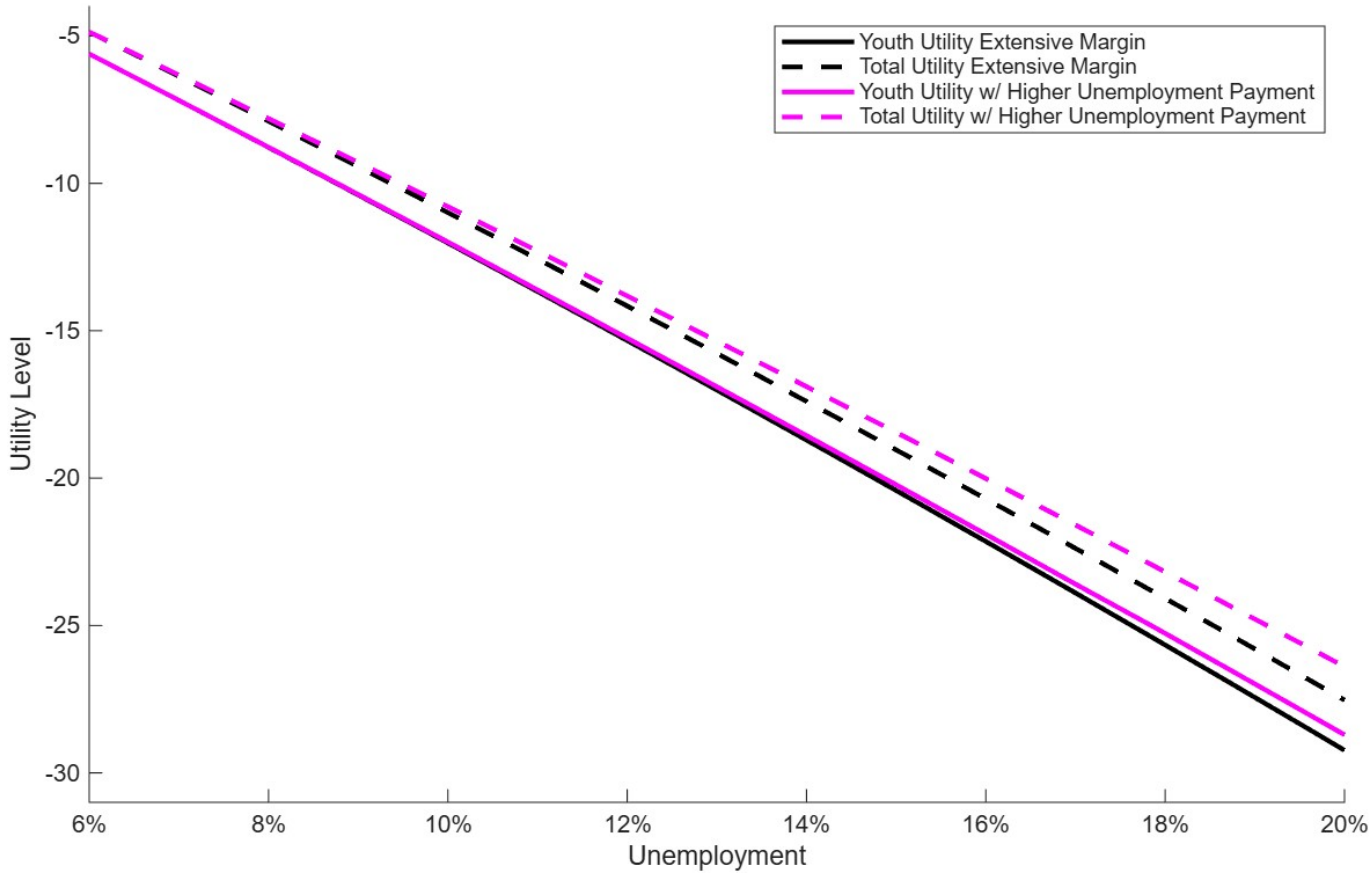
**Benefit of adding Countercyclical Transfer Payments**

- Adverse productivity shocks lower output, income, and utility.
- Raising social security and unemployment transfers mitigates the decline in household utility (vs. the fixed transfer payment benchmark).

**What The Payments Are**

- Transfers are inversely proportional to output: $T_{0,shock} = T_{0,ss} \times \frac{Y_{ss}}{Y_{shock}}$.

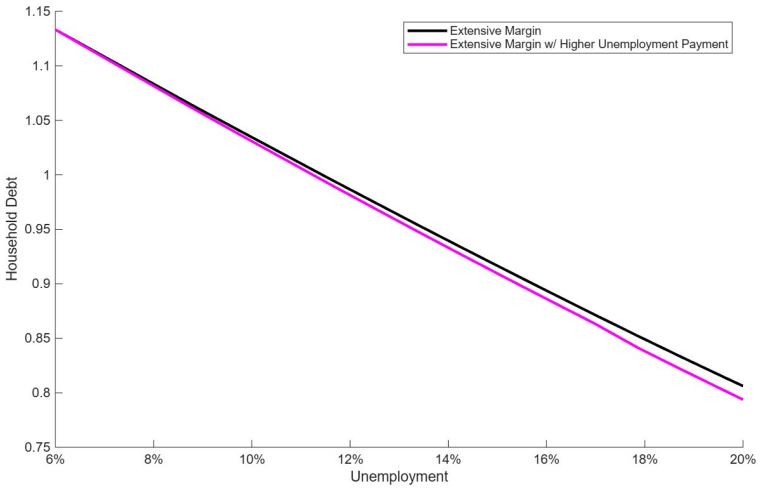
# Utility Increase Transfer Payments



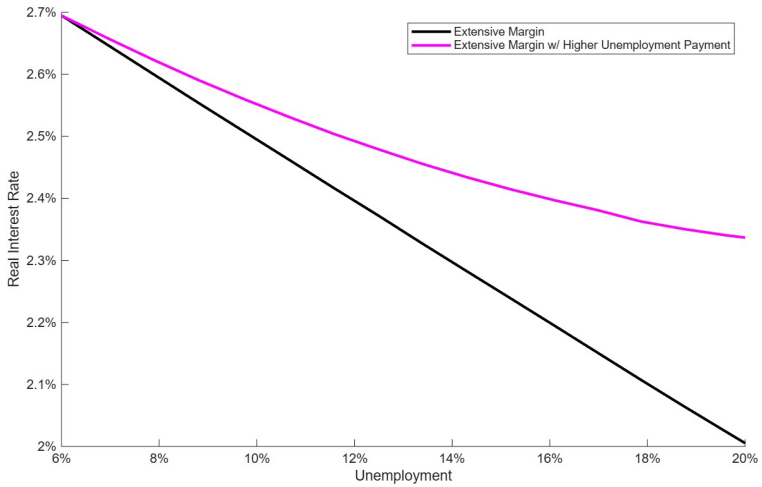

*Notes*: Increasing transfer payments increases utility.

Loanable Funds Market

# Comparative Statics Loanable Funds with Transfers



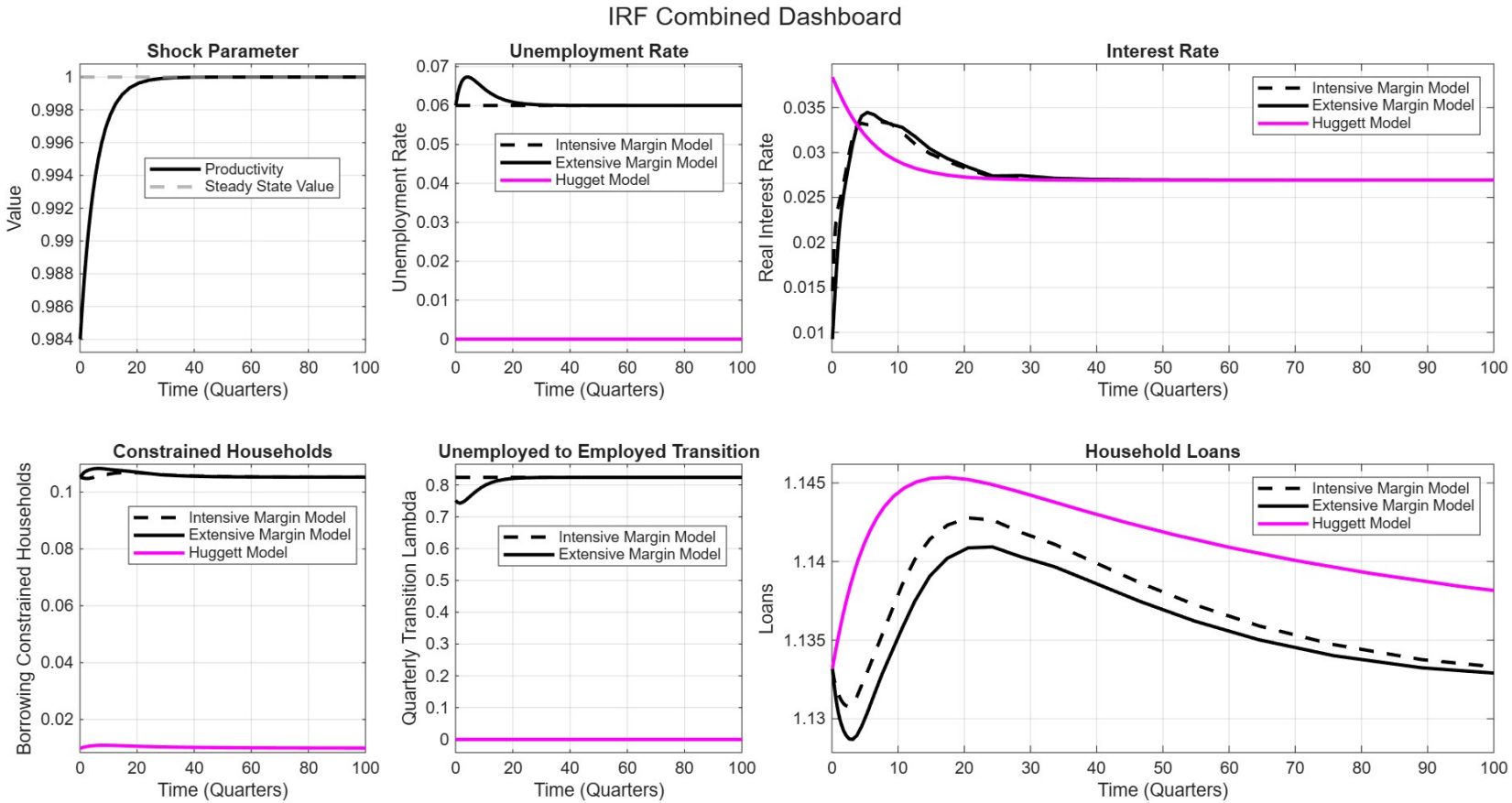

*Notes*: Counter cyclical transfers decrease the quantity of loans.




*Notes*: Real interest rates stop declining.

Return

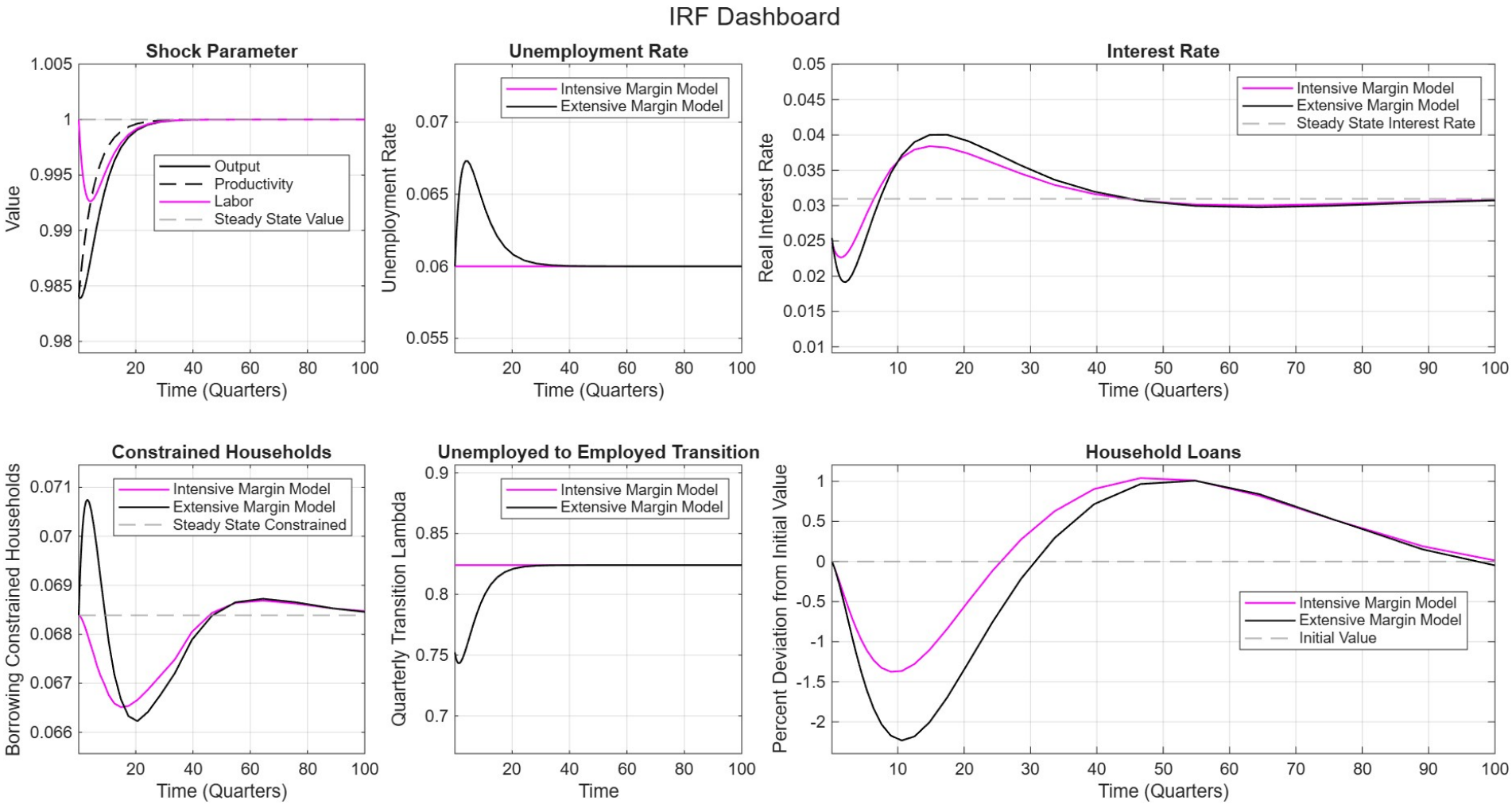
# Temporary Negative Productivity Shock Full Dashboard




*Notes*: Large standouts are the interest rate paths with the Extensive Margin Model having the lowest bottom before both spike above the Huggett Model. Both the Extensive and Intensive Models show an initial contraction in household debt before it increases.

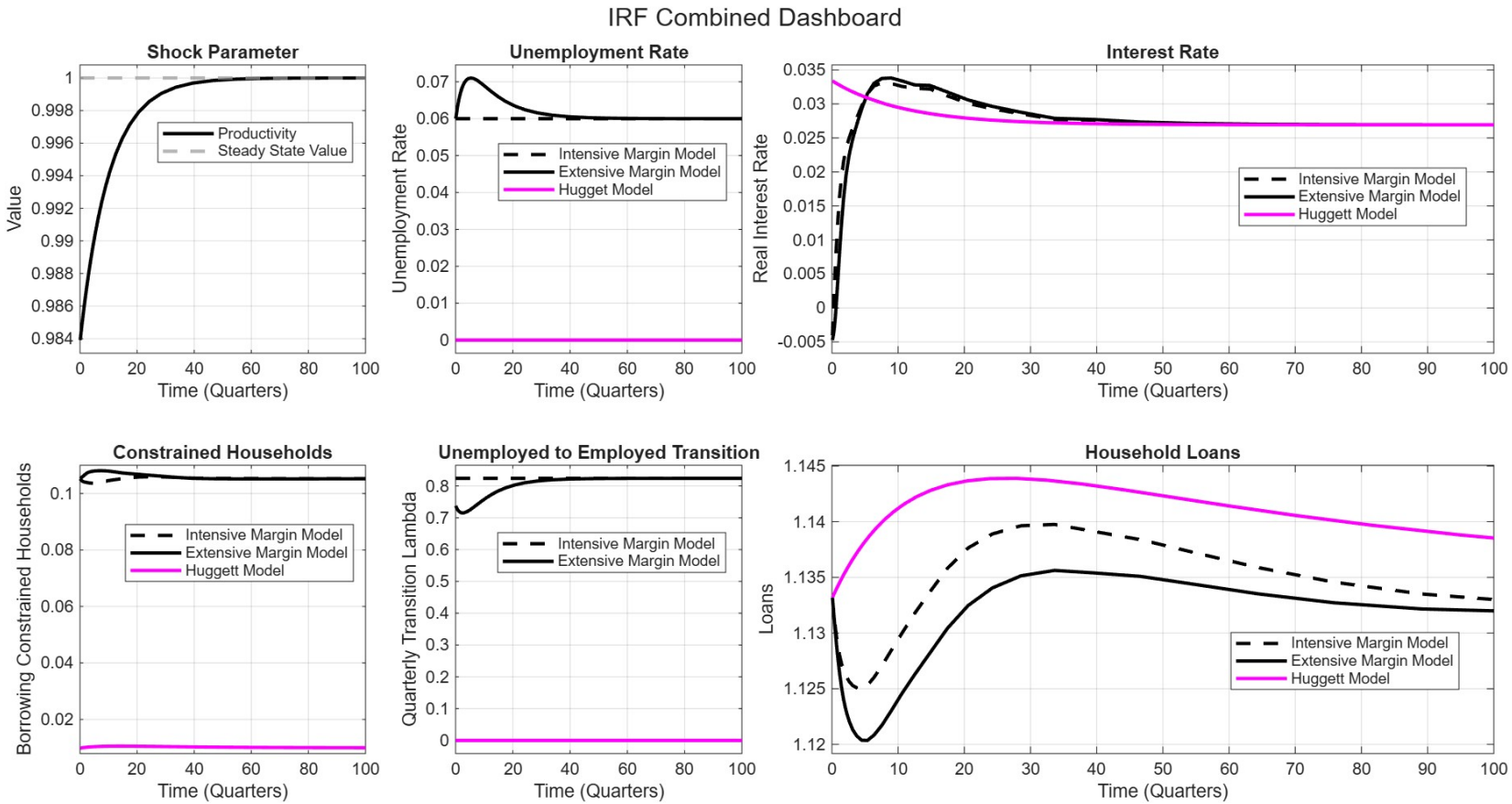
Return

# Including Government debt, deficits, and spending




*Notes*: G = .17, government debt to GDP is set to .7.

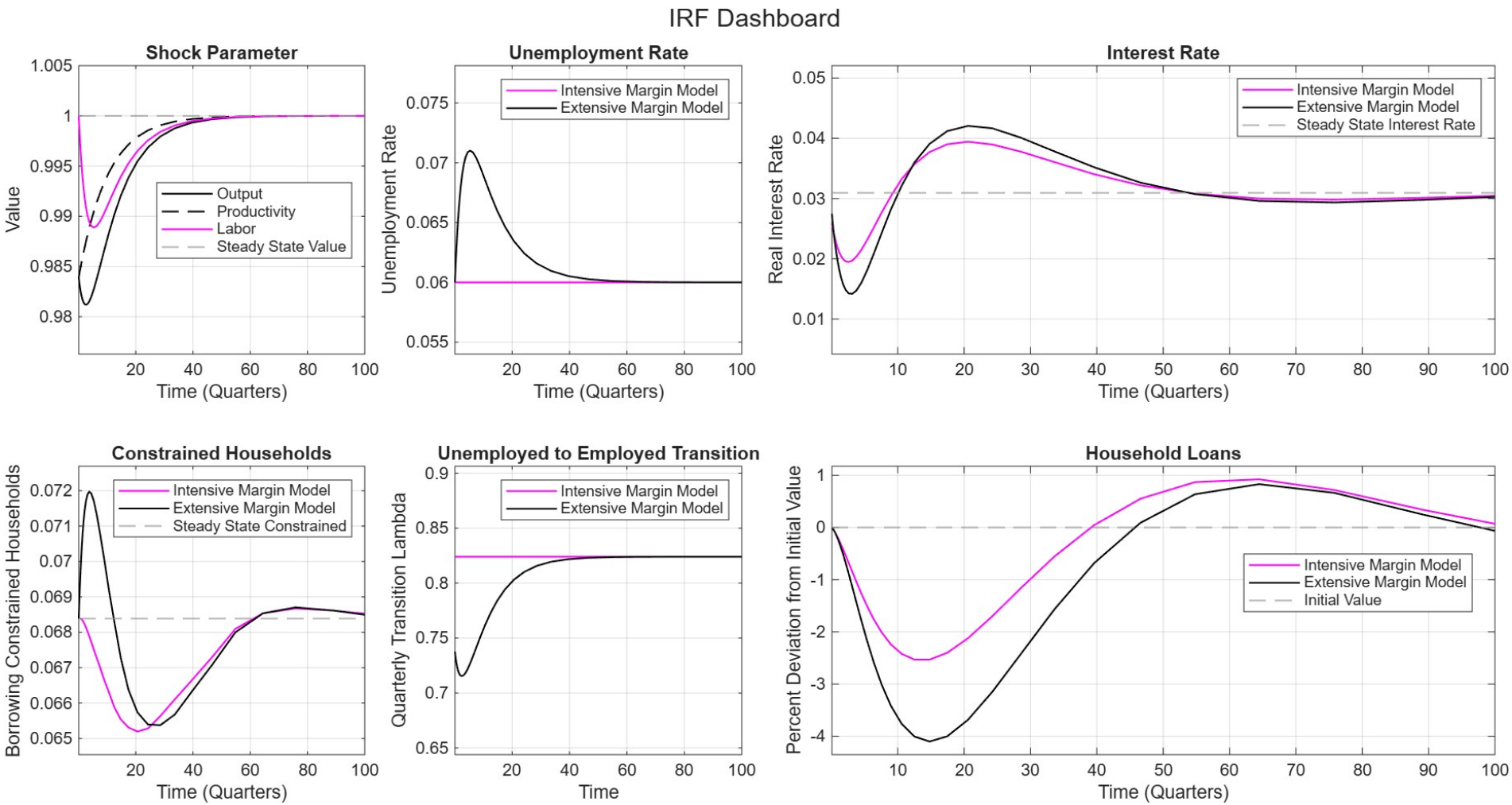
# Persistent Productivity Shock Full Dashboard




*Notes*: Increasing shock persistence to .9 creates the dynamics where the extensive margin does not overshoot the starting household loan value.

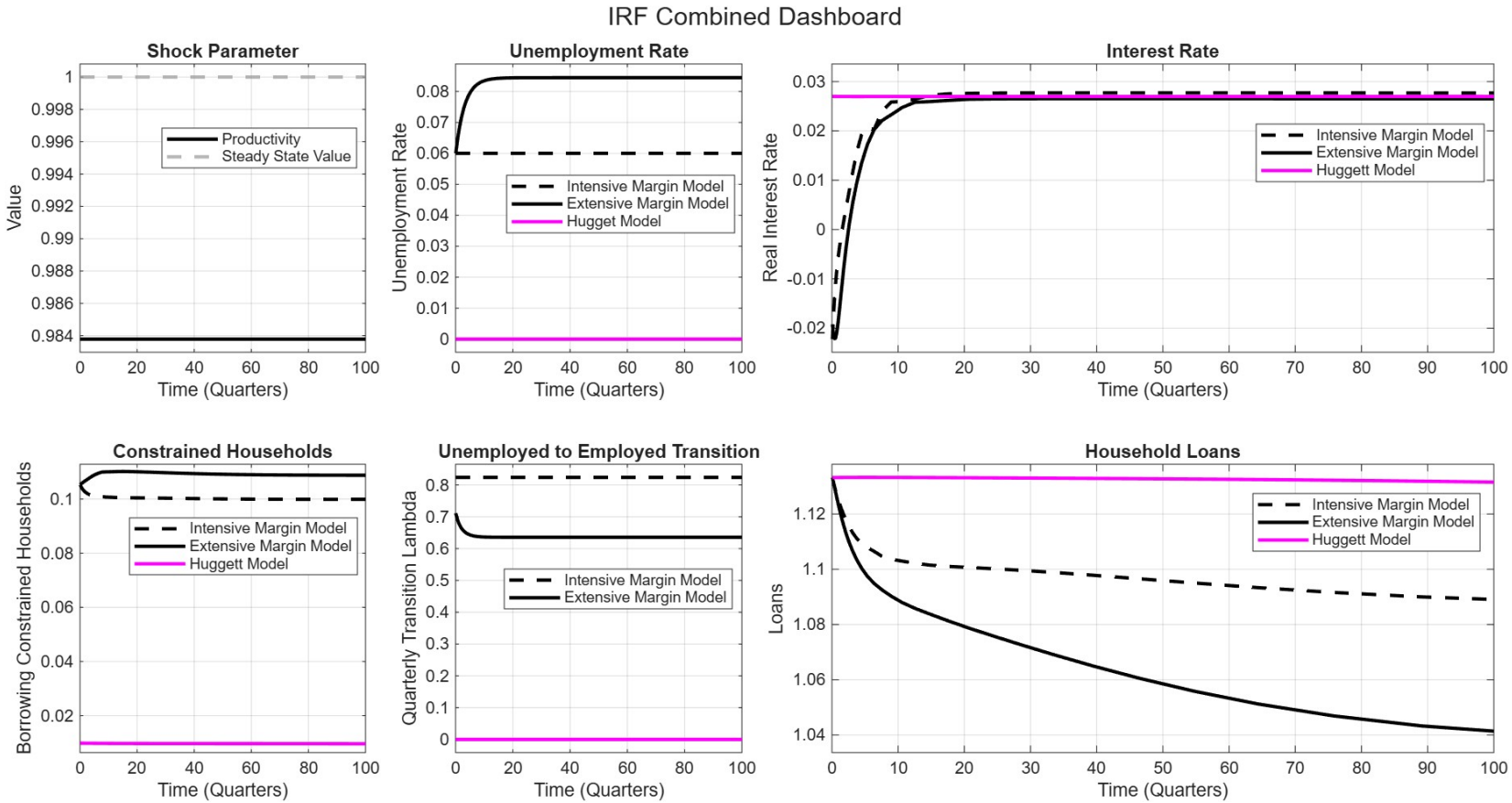
Return

# Including Government debt, deficits, and spending with higher persistence




*Notes*: G = .17, government debt to GDP is set to .7.

# Permanent Negative Productivity Shock Full Dashboard

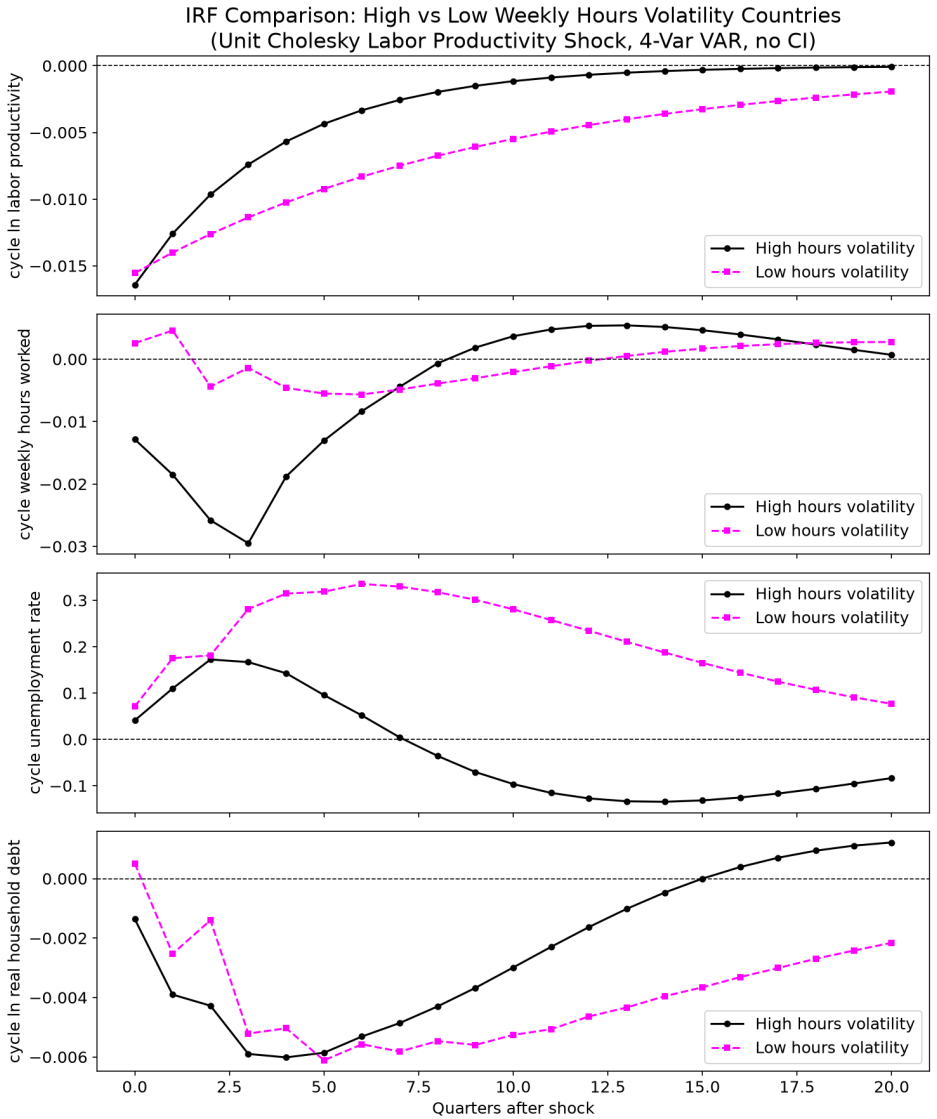



*Notes*: Difference in leverage response is large and interest rates have persistent differences.

Return

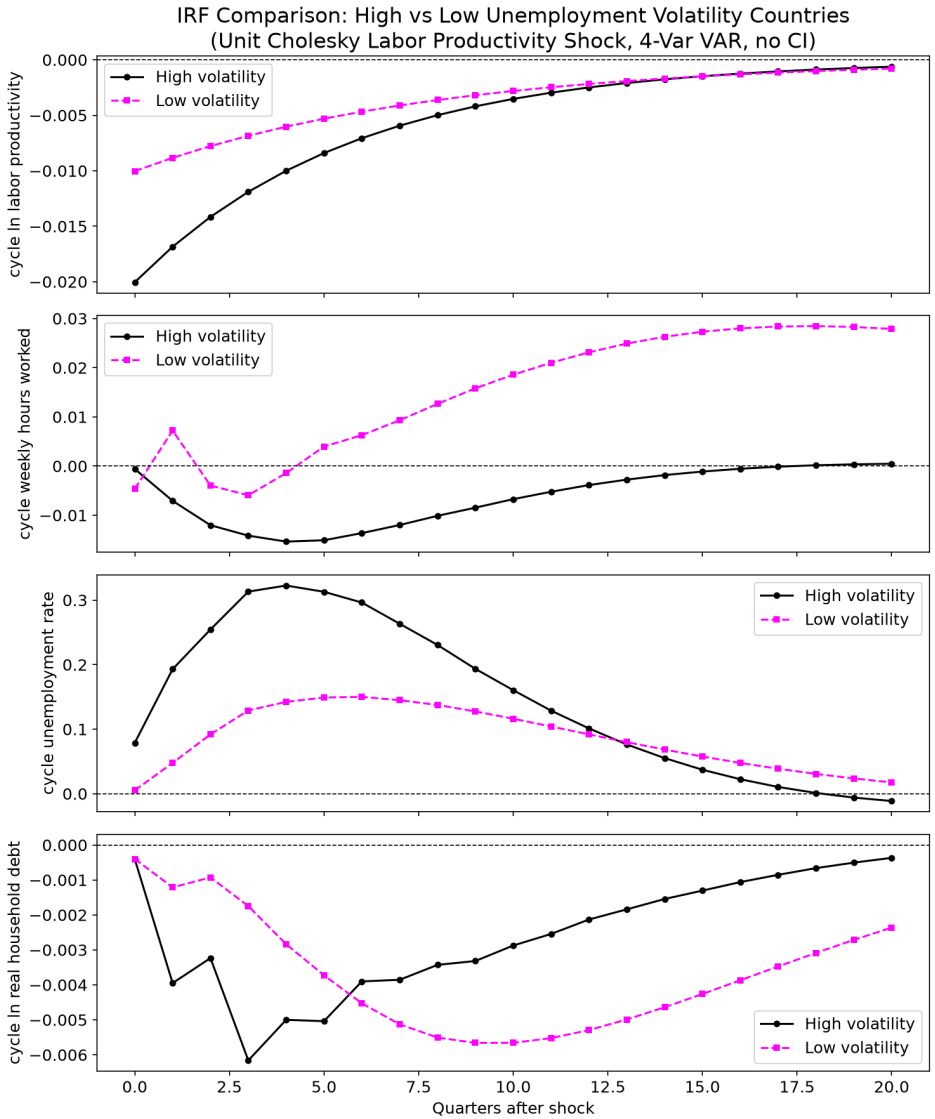
# VAR Weekly Hours Variance

- The countries are split based on variance in weekly hours worked.
- The median variance: 0.061210.
- High hours volatility countries (8): Austria, Czechia, Denmark, France, Ireland, Italy, Portugal, Spain.
- Low hours volatility countries (8): Belgium, Finland, Germany, Greece, Hungary, Norway, Poland, Sweden.



# VAR Unemployment Variance

- The countries are split based on the variance in the UR.
- Median variance: 2.1049.
- High UR volatility countries (8): Czechia, Greece, Hungary, Ireland, Italy, Poland, Portugal, and Spain.
- Low UR volatility countries (8): Austria, Belgium, Denmark, Finland, France, Germany, Norway, and Sweden.



# Interest Rate Optimization

## Levenberg-Marquardt

$$F_{aug} = [F; sqrt(\lambda_s) * Ps]; \tag{3}$$

$$J_{aug} = [J; sqrt(\lambda_s) * R]; \tag{4}$$

$$H = J'_{aug} * J_{aug} + \mu * eye(nT); \tag{5}$$

$$g = J'_{aug} * F_{aug}; \tag{6}$$

$$dx = -H^{-1} * G; \tag{7}$$